# Cross-Domain Empirical Evidence for $\Phi = I \cdot \rho - \alpha \cdot S$ as an Indicator of Operational Regime

Shawn Barnicle

May 2026

**Abstract**

We report the cross-substrate recurrence of a three-coordinate scalar, $\Phi = I \cdot \rho - \alpha \cdot S$, as a bounded empirical indicator of operational regime across tested mechanical, aerospace, electrical, geophysical, quantum, computational, biological, and large-language-model substrates. The parameters $\alpha = 0.1$ and $\Phi_c = 0.25$ for engineered-cluster substrates were fixed in production code on 2025-10-21 and reused unchanged in three subsequent repositories through 2025-12-31. A later observational LLM repository reused both values in code on 2026-01-27, with the threshold logged only and not used to gate execution. A USPTO provisional patent application filed on 2025-11-20 provides disclosure evidence independent of git history. The evidence takes three forms. Regime-separation evidence is reported for mechanical bearings, aerospace turbofans, electrical grids, geophysical cases, quantum qubits and two-qubit gates, and computational neural-network architectures: $\Phi$ falls below the applicable reference threshold in documented failing or critical cases and remains above it in the tested stable controls, where available, using domain-specific implementations of $I$, $\rho$, and $S$. Scope-boundary evidence is reported for biological cardiac data: under a matched protocol, $\Phi$ separates arrhythmia transition-positive windows from normal-symbol windows (AUC = 0.9148) but does not separate atrial fibrillation from non-AFIB rhythm (AUC = 0.5556), establishing a transition-versus-stable-state scope bound. Observational structural-role evidence is reported for large language models: $\Phi$ supports the substrate-shared structural role of the three coordinates without threshold-gated regime decisions. Most evidence is retrospective. A preregistered forward test on a mechanical system outside the original scalar validation set produced end-of-life $\Phi = -0.342$, satisfying the preregistered criterion $\Phi < 0.25$ under a commitment chain documented by git history. The pattern is not claimed to generalize beyond the tested substrates, and documented limitations and counterexamples are reported in the manuscript. Every numerical claim traces to a code-output artifact at a pinned commit hash in repositories released with this paper.

## 1 Introduction

Monitoring systems for approaching failure is a problem with substrate-specific traditions. Bearings are monitored by vibration spectra. Turbofan engines are monitored by engine-health models calibrated against run-to-failure corpora. Power grids are monitored by frequency-deviation detectors. Quantum qubits are monitored through calibration-based quality scores. Neural-network training is monitored by accuracy-based early-stopping heuristics. Cardiac patients are monitored by RR-interval classifiers and rhythm-policy annotation. Language models are monitored by perplexity-based or embedding-based drift detectors. Each of these tools is mature within its substrate, and each was built by a different community with different training conventions, different observables, and different notions of what counts as a signal of approaching failure.

This paper reports an empirical observation that sits alongside those substrate-specific tools. A single scalar, computed from three coordinates (identity, coherence, disorder), recurs across tested substrate families in three distinct forms of evidence: regime-separation evidence for mechanical, aerospace, electrical, geophysical, quantum, and computational substrates; scope-boundary evidence for biological cardiac data; and observational structural-role evidence for large-language-model outputs. The scalar is:

$$\Phi = I \cdot \rho - \alpha \cdot S$$

with $\alpha = 0.1$ and $\Phi_c = 0.25$ for engineered-cluster substrates. The coordinate observables differ by substrate; the structural roles do not (Section 2). The three forms of evidence are reported in Section 3 for the load-bearing substrates, Section 5 for the transition-versus-stable-state scope bound, and Section 3 for the LLM substrate-shared structural-role evidence.

Cross-domain empirical regularity reports face a standard reviewer concern: that the pattern reflects implicit per-domain parameter tuning dressed up as recurrence. This paper addresses that concern directly. The parameters $\alpha = 0.1$ and $\Phi_c = 0.25$ were fixed in production code on 2025-10-21 in a single repository and reused unchanged in three subsequent production repositories through 2025-12-31; a later observational LLM repository reused both values in code on 2026-01-27, with the threshold logged only and not used to gate execution. A USPTO provisional patent application filed on 2025-11-20 provides disclosure evidence for the parameter values independent of git history (Section 2). Specialized regimes (a 100-class CNN shifted threshold; biological cardiac with per-record threshold calibration; preliminary biological neural EEG calibration) appear in git history together with the evidence that motivated them, so these regimes were documented at first validation rather than retuned after the fact. A preregistered forward test on a mechanical system not in the original validation set (XJTU-SY Bearing2_1) produced end-of-life $\Phi = -0.342$, satisfying the preregistered criterion $\Phi < 0.25$ under a commitment chain documented by git history (Section 7).

The paper reports the pattern. It does not claim to explain it. The $(\alpha, \Phi_c)$ clustering observed across substrates is an empirical observation on the tested systems, not a derivation; what determines $(\alpha, \Phi_c)$ for a given system class is explicitly stated as an open question (Section 4). The scalar is not claimed to replace substrate-specific monitoring tools in their native tasks, and it is not claimed to generalize beyond the tested substrates. Independently of the regime-separation evidence, the same scalar has been reused across operationally distinct implementations in separately audited production repositories; this operational recurrence is reported in Section 6 as portfolio-level context and is explicitly non-load-bearing relative to the cross-substrate empirical claims.

The paper's contribution is the cross-substrate recurrence itself, bounded by the scope statements in Sections 4, 5, and 9. Each numerical claim traces to a code-output artifact at a pinned commit hash in repositories released with this paper (Section 11); independent verification is available by cloning the cited repositories at the cited commits and inspecting or running the cited artifacts and scripts.

Section 2 defines the scalar, the coordinate admissibility rules, and the chronology of parameter fixation. Section 3 reports the eight-substrate evidence. Section 4 reports the $(\alpha, \Phi_c)$ clustering. Section 5 reports the transition-versus-stable-state scope bound. Section 6 reports the operational recurrence across audited implementations. Section 7 reports the preregistered forward test. Section 9 documents failure modes and counterexamples. Section 10 positions the paper against adjacent literature. Section 11 provides the provenance manifest.

# 2 The Scalar and Its Coordinates

This section defines the scalar, the admissibility rules that govern what counts as each coordinate, the unit-of-analysis rule that governs how the scalar is reported across substrates, the repository-by-repository implementation of the three coordinates, and the chronology of parameter fixation that establishes the scalar's parameters were locked before cross-domain validation.

## 2.1 The Scalar Form

The scalar is:

$$\Phi = I \cdot \rho - \alpha \cdot S$$

with parameters $\alpha = 0.1$ and, for engineered-cluster substrates, $\Phi_c = 0.25$ as the reference threshold. Specialized or substrate-specific threshold regimes use different threshold conventions introduced at first validation, discussed in Section 4.

## 2.2 Admissibility Rules

All evidence reported in this paper expresses $I$, $\rho$, and $S$ through domain-appropriate observables that satisfy the following admissibility rules. The observables differ by substrate; their structural roles do not.

For a quantity to count as $I$: it must be an order-preservation or baseline-preservation coordinate. It measures how much of a reference structural signal is retained under the tested perturbation.

For a quantity to count as $\rho$: it must quantify temporal continuity, persistence across sequence, or sequential coherence. In batch-inference cases with no temporal structure, $\rho = 1.0$ is the documented default.

For a quantity to count as $S$: it must quantify disorder, dispersion, variability, or output diversity.

A coordinate implementation that does not satisfy the rule for its role is not admissible as evidence for this scalar.

## 2.3 Unit of Analysis Rule

The unit of analysis differs by substrate. This paper does not pool heterogeneous units into one grand statistical sample. Cross-domain recurrence is reported across heterogeneous unit types with per-domain counts. The substrate-specific units used in the evidence are: physical systems (mechanical bearings, aerospace turbofans), event windows (electrical grids, geophysical strainmeter cases), foreshock catalog sequences (geophysical

Tohoku case), individual qubits and two-qubit gate pairs (quantum), training architectures (computational neural networks), patient-windowed records (biological cardiac), and model instances under matched prompt suites (large language models).

## 2.4 Repository-by-Repository Coordinate Implementation

Table 1 reports the coordinate implementations used in each repository that contributes evidence to this paper. Each implementation satisfies the admissibility rules of Section 2; the observables differ by substrate but the structural role of each coordinate is preserved.

| **Repository (domain)** | **$I$ implementation** | **$\rho$ implementation** | **$S$ implementation** |
|---|---|---|---|
| -thermodynamic-stability-prediction (bearings) | baseline_RMS / current_RMS | Autocorr. lag 1 of RMS series | Shannon entropy (bits), 50 bins |
| -thermodynamic-stability-prediction (turbofans) | Per-sensor baseline_std / current_std, averaged across valid sensors | Per-sensor autocorr. lag 1, averaged | Per-sensor Shannon entropy (10 bins), averaged |
| -thermodynamic-stability-prediction (grids) | baseline_freq_dev_std / test_freq_dev_std | Autocorr. lag 1 of test window (last 1000 pts) | Shannon entropy (50 bins) of test window |
| -thermodynamic-stability-prediction (geophysical Donna Lea) | min(baseline_mean / current_mean, 1.0) | Autocorr. lag 1 of test window | Shannon entropy of test window |
| -thermodynamic-stability-prediction (geophysical foreshock) | Magnitude-accumulation baseline-vs-degraded split on 61-event catalog | Catalog-level autocorr. ($-0.328$ observed) | Shannon entropy of magnitude distribution |
| quantum-phi-validation | (fidelity $-$ 0.50)/0.50 | $T_2/T_1$ coherence-time ratio | Readout error (entropy proxy) |
| phi-controller (NN) | (acc $-$ random)/(1 $-$ random) | Autocorr. lag 1 of per-epoch accuracy | $S_{\text{norm}} = S(\text{CM})/\log_2(n^2)$ |
| phi-bio-stability (cardiac) | baseline_std_RR / current_std_RR | Autocorr. lag 1 of RR-interval series | Shannon entropy of RR-interval distribution |
| llm-phi-stability | $I_{\text{drift}}$ = cosine sim. of two condition mean embeddings, clamped to $[0, 1]$; each mean is avg. of $L^2$-norm. per-sample embeddings across prompts and samples, then $L^2$-norm. In drift format: baseline $I = 1.0$, comparison $I = I_{\text{drift}}$ | Cosine between successive window means (Test 4); 1.0 for batch tests | Upper-triangular pairwise cosine distance among samples, averaged and halved, clamped to $[0, 1]$ |

Table 1: Repository-by-repository implementation of the three coordinates. Each row satisfies the admissibility rules in Section 2. The observables differ across substrates; the structural role of each coordinate is preserved across all rows. LLM implementation verified against `llm-phi-stability/core/phi_llm.py` at commit `16fef3d` (functions `compute_embeddings_from_outputs`, `mean_embedding`, `cosine_sim01`, `compute_two_condition_phi`; lines 45–193).

## 2.5 Chronology of Parameter Fixation

The parameters $\alpha = 0.1$ and $\Phi_c = 0.25$ were first fixed in a single production repository on 2025-10-21 and reused unchanged in three subsequent production repositories through 2025-12-31; the LLM repository later reused both values in code on 2026-01-27, with the threshold logged only and not used to gate execution. Three specialized-regime cases appear at specific later commits, each documented at the time of that substrate's first validation; the biological neural (EEG) case is reported as preliminary.

Exploratory work on adaptive-threshold variants preceded the first production fixation on 2025-10-21. The chronology reported here cites only production repositories, defined as repositories whose initial commit established the canonical scalar formula with $\alpha = 0.1$ and $\Phi_c = 0.25$ in executable code; it does not attempt to reconstruct earlier exploratory sandbox work.

**First production fixation.** The earliest git-tracked commit containing $\alpha = 0.1$ and $\Phi_c = 0.25$ together in the full scalar formula is `system-degradation-framework` commit `5ec267f` dated 2025-10-21, file `scripts/test_thermodynamic_law.py`, containing the function signature `calculate_phi(identity, autocorr, entropy, alpha=0.1)` and the threshold decision `elif Phi_degraded < 0.25`. At the same commit, the repository contained other scripts using different $\alpha$ values for different formulas ($\alpha = 0.034$ for a battery-oriented variant, $\alpha = 0.15$ for an earlier bearing variant); the $\alpha = 0.1$ value specifically in the full scalar $\Phi = I \cdot \rho - \alpha \cdot S$ is the one that propagated forward into all subsequent production repositories.

**External disclosure evidence.** A USPTO provisional patent application was filed on 2025-11-20 with source code from `system-degradation-framework` as the disclosed material. The filing date provides disclosure evidence independent of git history that $\alpha = 0.1$ and $\Phi_c = 0.25$ were disclosed by 2025-11-20, before the subsequent cross-domain validation sequence reported here. The exact provisional-application identifier is reported in Section 11.

**Reused unchanged.** Three subsequent production repositories reused $\alpha = 0.1$ and $\Phi_c = 0.25$ unchanged through 2025-12-31, each establishing the parameter values verbatim in the initial production commit of the repository's scalar computation: `-thermodynamic-stability-prediction` at `a76ba47` (2025-11-25, `core/phi_calculator.py`); `phi-controller` at `d0554226` (2025-12-05, `core/phi_calculator.py` function defaults); and `quantum-phi-validation` at `a50f438` (2025-12-31, module-level constants `ALPHA = 0.1` and `THRESHOLD = 0.25` in three initial test files). The `llm-phi-stability` repository later reused both values at `d1d3f61` (2026-01-27, `ALPHA_DEFAULT = 0.1` and `THRESHOLD_DEFAULT = 0.25`), with the threshold logged only and not gating execution in this observational-only substrate. Separately, `phi-bio-stability` at `a151789` (2026-02-02) retained $\alpha = 0.1$ from the engineered-cluster anchor for the cardiac substrate while replacing the fixed $\Phi_c = 0.25$ with per-record threshold calibration at target false-positive rate 0.01 on NORMAL windows; the cardiac regime is therefore classified in the chronology as a specialized-regime case, not as reused-unchanged.

**Specialized regimes at first validation.** Three specialized-regime cases appear in the chronology, each documented at the time of that substrate's first validation and not retuned afterward; the EEG case is reported as preliminary. The specialized computational 100-class CNN case was introduced at `phi-controller` commit `a3e8ba2` (2025-12-10) with $\Phi_c = 0.082$ operational and $\Phi_c = 0.084$ as the edge-case architecture's exact $\Phi$, documented in `experiments/OUTPUTS.md`. The biological cardiac case was introduced at `phi-bio-stability` commit `a151789` (2026-02-02) replacing a fixed $\Phi_c$ with per-record threshold calibration at target false-positive rate 0.01 on NORMAL windows. The biological neural (EEG) exploratory case was introduced at `phi-bio-stability` commit `fa04607` (2026-02-04) with $\alpha \approx 0.55$ and patient-specific calibration required; this EEG evidence is reported as preliminary in Section 4 and is not included in the consolidated evidence table.

Table 2 consolidates the chronology into a compact reference. Each row traces to a specific commit hash in the cited repository, with the code expression establishing the parameter value at that commit listed in the provenance manifest of Section 11.

| Date | Repository / External anchor | Commit | $\alpha$ | $\Phi_c$ | Status |
|---|---|---|---|---|---|
| 2025-10-21 | `system-degradation-framework` | `5ec267f` | 0.1 | 0.25 | First-fixed (origin) |
| 2025-11-20 | USPTO provisional filing | external | 0.1 | 0.25 | Disclosure evidence |
| 2025-11-25 | `-thermodynamic-stability-prediction` | `a76ba47` | 0.1 | 0.25 | Reused unchanged |
| 2025-12-05 | `phi-controller` | `d0554226` | 0.1 | 0.25 | Reused unchanged |
| 2025-12-10 | `phi-controller` (100-class CNN) | `a3e8ba2` | 0.1 | 0.082 | Shifted at first validation |
| 2025-12-31 | `quantum-phi-validation` | `a50f438` | 0.1 | 0.25 | Reused unchanged |
| 2026-01-27 | `llm-phi-stability` | `d1d3f61` | 0.1 | 0.25 (ref. only) | Reused unchanged (not gating) |
| 2026-02-02 | `phi-bio-stability` (cardiac) | `a151789` | 0.1 | per-record FPR = 0.01 | Shifted at first validation |
| 2026-02-04 | `phi-bio-stability` (EEG) | `fa04607` | $\approx$ 0.55 | patient-specific | Shifted, preliminary |

Table 2: Chronology of scalar parameter fixation. The parameters $\alpha = 0.1$ and $\Phi_c = 0.25$ were first fixed in `system-degradation-framework` on 2025-10-21 and reused unchanged in three subsequent production repositories through 2025-12-31; a fourth later repository, `llm-phi-stability`, reused both values in code on 2026-01-27 with the threshold logged only and not gating execution. Three specialized-regime cases (100-class CNN, biological cardiac, biological neural EEG) appear later in the chronology, each documented at first validation and not retuned. A USPTO provisional patent application filed on 2025-11-20 provides disclosure evidence independent of git history.

**What the chronology supports.** The chronology above supports three points. First, $\alpha = 0.1$ and $\Phi_c = 0.25$ were fixed in production code on 2025-10-21, thirty days before the USPTO provisional filing date on 2025-11-20. Second, between 2025-10-21 and 2025-12-31, the same two parameter values were present in the origin repository and reused unchanged across three additional production repositories spanning mechanical, aerospace, electrical, geophysical, quantum, and standard-class computational substrate instantiations; the LLM repository at 2026-01-27 later reused both values unchanged in code but with the threshold logged only and not gating execution. For the biological cardiac substrate, $\alpha = 0.1$ was carried forward unchanged from the engineered-cluster anchor, while the fixed $\Phi_c$ was replaced with per-record threshold calibration at first validation. Third, specialized-regime parameters appear in git history together with the evidence that motivated them, so the audit trail supports the narrower conclusion that these values were documented at first validation rather than introduced later in the manuscript to fit already-reported domain-specific outcomes. The code expressions establishing each row of the chronology are verbatim from the earliest production commit in each repository.

# 3 Cross-Domain Empirical Results

This section reports the empirical results that motivate the paper. Eight substrates are evaluated against the same scalar form $\Phi = I \cdot \rho - \alpha \cdot S$ with $\alpha = 0.1$ across all engineered-cluster substrates. The engineered-cluster reference threshold $\Phi_c = 0.25$ is used for mechanical bearings, aerospace turbofans, electrical power grids, geophysical monitoring substrates (Donna Lea strainmeter cases and the Tohoku foreshock catalog), quantum qubits, and computational neural-network architectures at standard class counts. Specialized or substrate-specific threshold regimes use different threshold conventions introduced at first validation and not retuned afterward: the computational 100-class CNN case uses $\Phi_c = 0.082$ operational; the biological cardiac case uses per-record threshold calibration at target false-positive rate 0.01 on NORMAL windows rather than a fixed $\Phi_c$; the large-language-model case uses $\Phi_c = 0.25$ as a logged reference value only, with no threshold-gated pass/fail decision. Per-domain unit counts, datasets, and computed $\Phi$ values are reported in the consolidated evidence table at the end of this section. Per the unit-of-analysis rule, the eight substrates are not pooled into a single statistical sample; cross-substrate recurrence is reported across heterogeneous unit types, with per-domain counts reported for each substrate.

The mechanical, aerospace, electrical, geophysical, quantum, and computational substrate families are presented as load-bearing regime-separation evidence. The biological cardiac substrate is presented as scope-boundary evidence for the transition-versus-stable-state distinction: the positive arrhythmia-transition case appears in this section, and the negative AFib alternative-rhythm case appears in Section 5. The large-language-model substrate is presented as supporting evidence for the substrate-shared structural role of the three coordinates $I$, $\rho$, and $S$, rather than as load-bearing regime-separation evidence. The LLM tests are observational; no $\Phi$ threshold gates execution in those tests.

## 3.1 Ground-Truth Label Protocol

Table 3 documents how stable, critical, and failing labels were assigned in each domain. The protocols vary across substrates but share a common discipline: labels are assigned from external sources independent of the $\Phi$ computation, and no label is set after $\Phi$ is observed.

| Domain | Stable label source | Critical/failing label source | Window definition | Direction |
|---|---|---|---|---|
| Mechanical (bearings) | N/A (run-to-failure cases only) | XJTU-SY dataset documented end-of-life | Final vibration window | Retrospective |
| Aerospace (turbofans) | N/A (run-to-failure cases only) | NASA C-MAPSS documented engine failure cycle | Final operational cycle | Retrospective |
| Electrical (grids) | Germany Sept 2019 normal operation window | UK Aug 9, 2019 pre-blackout window (blackout at 16:52 UTC) | Multi-day baseline/test paired windows (UK: Aug 1-8 baseline, Aug 9 pre-blackout test; Germany: Sept 1-7 baseline, Sept 8-14 test) | Retrospective |
| Geophysical (Donna Lea same-sensor) | Donna Lea 2010 quiet year strain window | Donna Lea pre-event windows for Parkfield M6.0 and San Simeon M6.5 | Pre-event strain observation windows | Retrospective |
| Geophysical (foreshock catalog) | N/A (catalog comparison only) | USGS foreshock catalog for Tohoku M9.1 (61 events) | Catalog baseline-vs-degraded split | Retrospective |
| Quantum | Non-dead qubits in the IBM calibration snapshot across 3 backends (ibm_fez, ibm_torino, ibm_marrakesh) | Dead qubits flagged by IBM calibration as failing measurement on the same snapshot | Daily calibration snapshot | Same-day |
| Computational (NN) | Architectures reaching class-count-appropriate post-training accuracy threshold defined in the corresponding test outputs | Architectures failing to reach the post-training accuracy threshold at full training | Full training trajectory | Concurrent |
| Biological (cardiac) | MIT-BIH NORMAL beat windows | MIT-BIH ARRHYTHMIA beat windows | 60-second beat windows | Retrospective |
| LLM (observational, no regime label) | Baseline configuration mean embedding | Comparison-configuration mean embedding under matched prompts (not a failure label; observational condition contrast) | Per-prompt sample sets | Paired comparison |

Table 3: Ground-truth label protocol per domain. For the load-bearing empirical substrates (bearings, turbofans, grids, geophysical Donna Lea, geophysical foreshock catalog, quantum, and computational as regime-separation evidence; cardiac as transition-detection evidence), labels are assigned from external sources independent of the $\Phi$ computation: retrospective labels mean the failure outcome was already documented in the dataset before $\Phi$ was computed. The geophysical domain is split into two rows because the Donna Lea strainmeter cases (Parkfield, San Simeon, 2010 quiet year) are same-sensor time-series comparisons while the Tohoku case uses the USGS foreshock catalog with magnitude-accumulation comparison; the two are methodologically distinct. For the computational (NN) row, viability labels are assigned from final post-training accuracy against a class-count-appropriate threshold defined in the test outputs, independent of $\Phi$'s kill decision. For the quantum row, the hard external label is the dead-qubit designation reported by IBM calibration; intermediate categories (MARGINAL, GOOD) are the paper's own $\Phi$-based cohorting and are treated as the model's output, not ground truth. The LLM row is not a failure-labeling row; it is a paired-comparison row recording observational $\Phi$ values between baseline and comparison configurations under matched prompt suites. LLM evidence supports the substrate-shared structural-role observation and does not contribute regime-separation evidence.

## 3.2 Mechanical: Bearings

Ten bearings from the XJTU-SY Bearing Dataset (24) were evaluated against $\Phi = I \cdot \rho - \alpha \cdot S$ with $I$ = baseline_RMS/current_RMS, $\rho$ = autocorrelation of the RMS time series at lag 1, and $S$ = Shannon entropy in bits of the RMS distribution. All ten bearings are documented run-to-failure cases. End-of-life $\Phi$ values fall in the range $-0.370$ to $-0.003$, all below the engineered-cluster reference $\Phi_c = 0.25$. Per the regime classification used in this paper, all ten bearings end in the collapse regime ($\Phi < 0$). The values closest to zero (Bearing1_4 at $-0.003$) are near the collapse/critical boundary; the others are well inside collapse. The reference implementation is `paper_validation/scripts/compute_bearing_components.py` in the `-thermodynamic-stability-prediction` repository at commit `2e2099c`. The preregistered Bearing2_1 cold test is reported separately in Section 7 and is not included in this ten-bearing retrospective range.

## 3.3 Aerospace: Turbofans

Ten engines from the NASA C-MAPSS Turbofan Engine Degradation Simulation Dataset (3) were evaluated against the same scalar with multi-sensor baseline-vs-current averaging excluding zero-variance channels in the $I$ computation. All ten engines are documented run-to-failure cases. End-of-life $\Phi$ values fall in the range 0.039 to 0.241, all below $\Phi_c = 0.25$. All ten engines end in the critical regime under this paper's classification. The reference implementation is `paper_validation/scripts/compute_turbofan_components.py` in the `-thermodynamic-stability-prediction` repository at commit `2e2099c`. Per-engine end-of-life values trace to `paper_validation/outputs/turbofans_components.json` at the same commit.

## 3.4 Electrical: Grids

Two power grids were evaluated as a paired comparison on the same instrument family using public grid frequency data (23). The UK National Grid August 9, 2019 blackout window produced $\Phi = 0.178$, in the critical regime. The Germany September 2019 normal-operation window on the same instrument family produced $\Phi = 0.401$, in the stable regime. The $I$ component is computed as baseline_frequency_deviation_std/current_frequency_deviation_std. The paired result on the same instrument family is the only paired stable-versus-critical comparison in the electrical domain of this paper. Canonical log files at commit `e36dcad` in the same repository document both Phi computations.

## 3.5 Geophysical: Strainmeter and Foreshock Catalog

The geophysical domain comprises two methodologically distinct substrate instantiations. The first is a same-sensor three-outcome validation on the Donna Lea borehole strainmeter near Parkfield, California. The pre-event window for the Parkfield M6.0 earthquake of September 28, 2004 produced $\Phi = 0.114$, in the critical regime. The pre-event window for the San Simeon M6.5 earthquake of December 22, 2003 produced $\Phi = 0.084$, in the critical regime. The 2010 quiet-year window on the same sensor produced $\Phi = 0.577$, in the stable regime. For these three Donna Lea cases, $I$ is computed as min(baseline_strain_mean/current_strain_mean, 1.0), $\rho$ as the autocorrelation at lag 1 of the test window, and $S$ as Shannon entropy in bits of the normalized test-window distribution. The same-sensor three-outcome design removes instrument-family confounds: one sensor produces two pre-event $\Phi$ values below $\Phi_c$ and one quiet-year $\Phi$ value well above $\Phi_c$.

The second substrate instantiation is a separate computation on the USGS foreshock catalog for the Tohoku M9.1 earthquake of March 11, 2011. Using a 61-event foreshock catalog with magnitude-accumulation comparison across the baseline-versus-degraded split, the Tohoku case produced $\Phi = -0.357$, in the collapse regime. This is a methodologically distinct computation from the Donna Lea strainmeter cases: the unit of analysis is a foreshock sequence rather than a strainmeter time series, the $\rho$ value is $-0.328$, reflecting catalog-level sequential anti-correlation rather than sensor-level persistence, and the implementation does not share the Donna Lea strainmeter pipeline. This use satisfies the $\rho$ admissibility rule as a sequential-coherence coordinate for the catalog ordering, not as a positive-persistence measure; the negative value contributes downward to $\Phi$ and is reported as part of the catalog-level collapse-regime signal rather than clipped away. The magnitude-accumulation baseline-versus-degraded split is used as the identity coordinate because it measures preservation of the reference catalog-accumulation structure under the degraded split, satisfying the baseline-preservation admissibility rule of Section 2. Both geophysical substrate instantiations satisfy the admissibility rules in Section 2. Canonical log files at commit `7ab6f2a` in the `-thermodynamic-stability-prediction` repository document all four computations.

### 3.6 Quantum: Qubits

Four hundred forty-five qubits across three IBM Quantum backends (`ibm_fez`, `ibm_torino`, `ibm_marrakesh`) were evaluated as one-day calibration snapshots. The $I$ component is computed as (fidelity $-$ 0.50)/0.50, $\rho$ as the $T_2/T_1$ coherence-time ratio (per the admissibility rule in Section 2, quantifying temporal persistence of the quantum state), and $S$ as readout error as an entropy proxy (quantifying measurement-outcome disorder). Of 445 qubits, 391 produced $\Phi \geq 0.25$ (GOOD), 49 produced $0 \leq \Phi < 0.25$ (MARGINAL), and 5 produced $\Phi < 0$ (dead). All five dead qubits separate cleanly below the $\Phi < 0$ collapse boundary. Across 1,004 two-qubit gates evaluated separately, the gate-pair group with $\min(\Phi) < 0.25$ shows a mean error rate 4.34$\times$ higher than the gate-pair group with $\min(\Phi) \geq 0.25$. The Pearson correlation between $\Phi$ and the published $T_2/T_1$ relaxation ratio across the 445 qubits is $r = 0.9458$, supporting the interpretation that $\Phi$ tracks an established calibration-quality signal. The reference implementation is `quantum-phi-validation/test_quantum_phi.py`, `test_quantum_phi_2qubit_gates.py`, and `test_quantum_phi_all_backends.py` at commit `a50f438` in the `quantum-phi-validation` repository. The $T_2/T_1$ correlation source is documented in `Outputs_MD/MASTER_VALIDATION_OUTPUT.md` at commit `21b191e` in the same repository. A single-day same-backend counterexample on `ibm_marrakesh` is documented in Section 9.

### 3.7 Computational: Neural-Network Architectures

The canonical `phi-controller` validation summary reports 660 architectures evaluated against the same scalar with $I = (\text{accuracy} - \text{random})/(1 - \text{random})$, $\rho$ = autocorrelation of the per-epoch accuracy history, and $S$ = normalized confusion-matrix entropy. The summary divides the 660 architectures into 445 MLP entries, 115 ten-class CNN entries, and 100 one-hundred-class CNN entries. Aggregate kill precision is 99.7% (2 false kills); category-level precision is 100% on MLPs, 98.3% on ten-class CNNs, and 100% on one-hundred-class CNNs at the optimized operational threshold $\Phi_c = 0.082$, with $\Phi = 0.084$ retained as the edge-case architecture's exact $\Phi$ value. Per-test kill precision ranges from 98.0% to 100.0% across the canonical test scripts. The reference implementation is `phi-controller/core/phi_calculator.py` at commit `d0554226`; the optimized 100-class threshold is introduced at commit `a3e8ba2` in the same repository. Two false kills on a CIFAR-10 CNN test (architectures at 60.6% and 61.5% accuracy killed at $\Phi_c = 0.25$) are documented in Section 9.

### 3.8 Biological: Cardiac Arrhythmia Detection

The MIT-BIH Arrhythmia Database (14) was evaluated with $I$ defined as baseline_std_RR divided by current_std_RR, $\rho$ as the autocorrelation of the RR-interval series at lag 1, and $S$ as the Shannon entropy of the RR-interval distribution. Of 48 total patient records in the database, 34 records were processed; 14 records were skipped due to insufficient pure NORMAL beat windows required for per-record threshold calibration at target false-positive rate 0.01. Across the 34 processed records, $\Phi$ achieved AUC = 0.9148, precision = 91.71%, recall = 72.02%, specificity = 94.65%, F1 = 0.8068, and MCC = 0.6936 for arrhythmia-window detection. The biological cardiac case uses a per-record threshold calibration rather than the engineered-cluster $\Phi_c$ = 0.25, with calibrated per-record thresholds clustering near $\Phi_c = 0$. This substrate-specific threshold characteristic is discussed in Section 4. The reference implementation is `phi-bio-stability/scripts/test_phi_ecg_mitdb.py` at commit `a151789`.

### 3.9 Large Language Models (Supporting Evidence)

Eight observational tests on three large-language-model architectures (TinyLlama 1.1B-Chat-v1.0 as primary, GPT-2 125M as cross-architecture comparison, TinyLlama 1.1B-intermediate-step-1431k-3T as fine-tuning-pair base) were evaluated against the same scalar. The identity component $I$ is computed by $L^2$-normalizing each per-sample output embedding, averaging the normalized embeddings across all prompts and samples within a condition, $L^2$-normalizing the resulting mean, and taking the cosine similarity between the two condition mean embeddings clamped to $[0, 1]$. The temporal coherence $\rho$ is computed as cosine similarity between successive window mean embeddings for temporal tests (Test 4), clamped to $[0, 1]$ and averaged across adjacent-window pairs; for single-snapshot tests (Tests 1, 2, 3, 5, 6, 7, 8), $\rho = 1.0$ by construction per the admissibility rule in Section 2 for batch inference without temporal structure. The disorder component $S$ is computed as the upper-triangular pairwise cosine distance among sample embeddings within each prompt, averaged across pairs and then across prompts, divided by 2 to map the cosine-distance range $[0, 2]$ into $[0, 1]$, and clamped to $[0, 1]$. The eight tests cover baseline reproducibility (Test 1), quality drift (Test 2), safety drift (Test 3), temporal stability (Test 4), temperature sensitivity (Test 5), cross-architecture comparison (Test 6), fine-tuning drift (Test 7), and adversarial-injection drift (Test 8).

The LLM tests are observational; the threshold $\Phi_c = 0.25$ appears in test code as a logged reference value but does not gate pass/fail in any test. The most informative single result is Test 6's cross-architecture comparison: TinyLlama 1.1B-Chat-v1.0 produced $\Phi = 0.973$ and GPT-2 (125M) produced $\Phi = 0.198$ on the same 20-prompt evaluation suite under the same embedding model, an absolute difference of 0.775 in $\Phi$. The cross-architecture $\Phi$ separation is presented as supporting evidence for the substrate-shared structural role of the three coordinates, not as load-bearing evidence for regime separation. The reference implementation is `llm-phi-stability/core/phi_llm.py` at commit `16fef3d`.

## 3.10 Consolidated Evidence Table

Tables 4 and 5 together form the consolidated evidence table required by the paper's control document. The table is presented in two parts for readability: Part A gives the coordinate definitions ($I$, $\rho$, $S$) per substrate; Part B gives the thresholds, result types, stable/transition ranges, and repositories. Every row traces to a specific code-output artifact (JSON file, markdown output, or log file as appropriate per repository) at a pinned commit hash. Per the unit-of-analysis rule, the System Count column reports the unit type appropriate to each substrate; counts are not pooled across rows. The geophysical domain is split into two rows reflecting the two methodologically distinct substrate instantiations described in Section 3. The biological cardiac and 100-class CNN entries use specialized or substrate-specific thresholds discussed in Section 4. The LLM row supports the substrate-shared structural-role observation rather than the regime-separation evidence.

| **Domain** | **Unit** | **Dataset / Protocol** | **System Count** | $I$ **Definition** | $\rho$ **Definition** | $S$ **Definition** |
|---|---|---|---|---|---|---|
| Mechanical | Physical system | XJTU-SY run-to-failure | 10 bearings | baseline_RMS / current_RMS | Autocorr. at lag 1 of RMS series | Shannon entropy (bits) of RMS distribution |
| Aerospace | Physical system | NASA C-MAPSS train_FD001 | 10 engines | Per-sensor baseline_std / current_std, averaged | Per-sensor autocorr. lag 1, averaged | Per-sensor Shannon entropy (10 bins), averaged |
| Electrical | Event window | National grid frequency data | 2 grids (paired) | baseline_std / test_std of freq deviation | Autocorr. lag 1 of test window | Shannon entropy (50 bins) of test window |
| Geophysical (Donna Lea) | Event window | USGS Donna Lea strainmeter | 3 cases (same sensor) | min(baseline_mean / current_mean, 1.0) | Autocorr. lag 1 of test window | Shannon entropy of test-window distribution |
| Geophysical (foreshock) | Foreshock catalog sequence | USGS foreshock catalog, Tohoku M9.1 | 1 event (61-event catalog) | Magnitude-accumulation baseline-vs-degraded split | Catalog-level autocorr. ($-0.328$) | Shannon entropy of magnitude distribution |
| Quantum | Qubit (and gate pair) | IBM calibration data, 3 backends | 445 qubits; 1004 gates | (fidelity − 0.50)/0.50 | $T_2/T_1$ coherence-time ratio | Readout error (entropy proxy) |
| Computational | NN architecture | MNIST, CIFAR-10/100, etc. | 660 architectures (445 MLPs, 115 CNN-10, 100 CNN-100) | (acc − random)/(1 − random) | Autocorr. lag 1 of per-epoch accuracy | Norm. confusion-matrix entropy $S/\log_2(n^2)$ |
| Biological cardiac | Patient record (60s) | MIT-BIH Arrhythmia Database | 34 of 48 processed | baseline_std_RR / current_std_RR | Autocorr. at lag 1 of RR-interval series | Shannon entropy of RR-interval distribution |
| LLM (structural-role support) | Model-under-prompt-suite | 8 tests, 3 models | 8 tests | Cosine sim. of $L^2$-norm. mean output embeddings | Cosine between successive window means; 1.0 for single-snapshot | Pairwise cosine distance among samples, /2, clamped |

Table 4: Consolidated evidence per domain, Part A: coordinate definitions. Continued in Table 5 with thresholds, result types, and repository references. Per the unit-of-analysis rule, System Count uses substrate-appropriate unit types and is not pooled across rows.

| **Domain** | $\alpha$ | $\Phi_c$ | **Primary Result Type** | **Stable / Reference** | **Failing / Transition** | **Repo** |
|---|---|---|---|---|---|---|
| Mechanical | 0.1 | 0.25 | Threshold separation (collapse) | N/A in dataset | $\Phi$ range $-0.370$ to $-0.003$ (all collapse) | -thermodynamic-stability-prediction |
| Aerospace | 0.1 | 0.25 | Threshold separation (critical) | N/A in dataset | $\Phi$ range 0.039 to 0.241 (all critical) | -thermodynamic-stability-prediction |
| Electrical | 0.1 | 0.25 | Paired stable/critical | Germany Sept 2019, $\Phi = 0.401$ | UK Aug 9 2019, $\Phi = 0.178$ | -thermodynamic-stability-prediction |
| Geophysical (Donna Lea) | 0.1 | 0.25 | Same-sensor three-outcome separation | 2010 quiet, $\Phi = 0.577$ | Parkfield $\Phi = 0.114$; San Simeon $\Phi = 0.084$ | -thermodynamic-stability-prediction |
| Geophysical (foreshock) | 0.1 | 0.25 | Threshold separation (collapse) | N/A (single-event) | Tohoku $\Phi = -0.357$ (collapse) | -thermodynamic-stability-prediction |
| Quantum | 0.1 | 0.25 | Dead-qubit threshold + gate-error discrim.; $r = 0.9458$ with $T_2/T_1$ | 391 qubits $\Phi \geq 0.25$ (GOOD) | 5 dead qubits $\Phi < 0$; 49 MARGINAL; $4.34\times$ gate-error ratio | quantum-phi-validation |
| Computational | 0.1 | 0.25 standard; 0.082 op. (100-class) | Kill-precision of trajectory-aware NN supervision | Architectures reaching post-training accuracy threshold | Aggregate 99.7% kill prec. (2 false kills); per-test 98.0–100.0% | phi-controller |
| Biological cardiac | 0.1 | Per-record (FPR = 0.01; cluster near 0) | Transition-detection scope-boundary evidence | NORMAL windows (TN = 531) | Arrhythmia: AUC 0.9148, MCC 0.6936 | phi-bio-stability |
| LLM (structural-role support) | 0.1 | 0.25 logged only; not gating | Observational substrate-shared structural-role evidence | Test 01 baseline $\Phi = 1.0$; Test 06 TinyLlama $\Phi = 0.973$ | Test 06 GPT-2 $\Phi = 0.198$ ($\Delta = 0.775$) | llm-phi-stability |

Table 5: Consolidated evidence per domain, Part B: thresholds, result types, and repositories. Continues Table 4. Every row traces to a code-output artifact (JSON, markdown, or log file as appropriate per repo) at a pinned commit hash. Specialized or substrate-specific thresholds (cardiac per-record, 100-class CNN $\Phi_c = 0.082$) are discussed in Section 4. The LLM row supports the substrate-shared structural-role observation and is not regime-separation evidence.

## 3.11 Cross-Domain Phi Distribution

Figure 1 visualizes the $\Phi$ values reported above. Each substrate is shown as a row; each system within a substrate appears as a single point colored by regime classification. The figure is the visual summary of the cross-domain recurrence reported in this section. The figure caption identifies the unit of analysis per row to preserve the unit-of-analysis discipline.

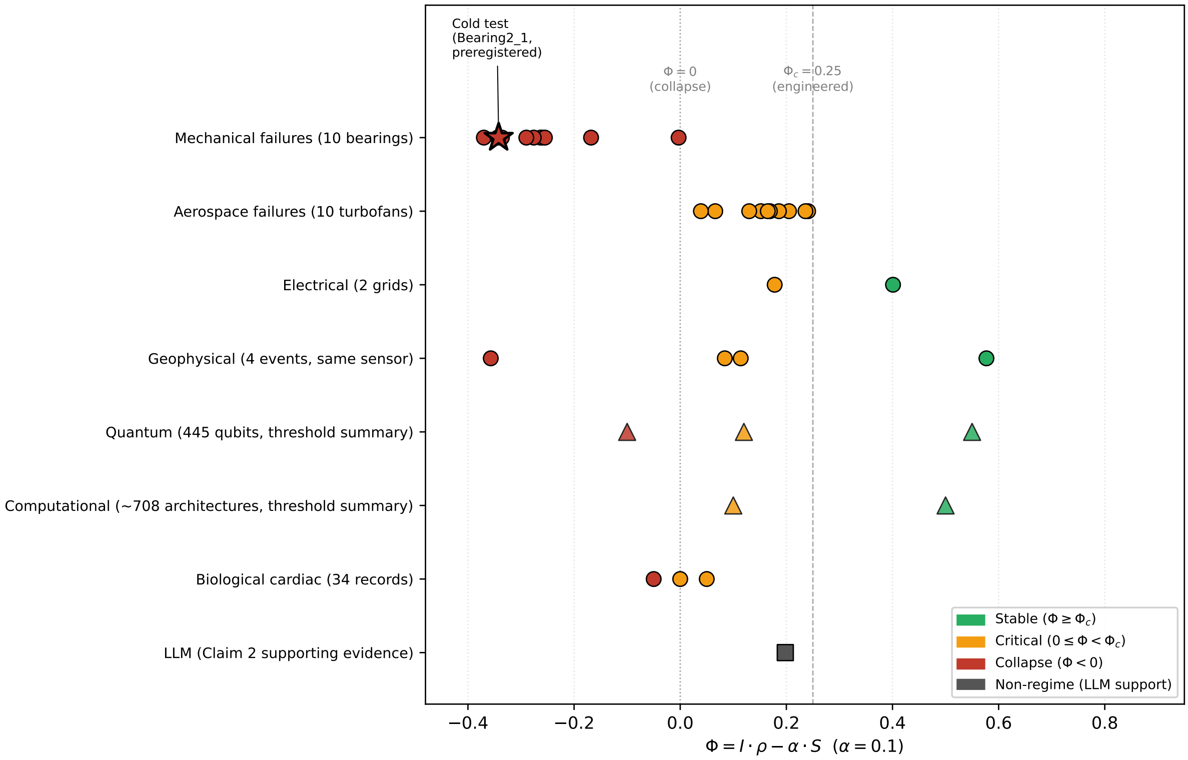


Figure 1: Cross-domain $\Phi$ distribution across the tested substrates. Each point is one unit of analysis (physical system, event window, qubit, NN architecture, patient record, or model-under-prompt-suite, per Tables 4 and 5). For the load-bearing regime-separation substrates, color encodes regime: stable ($\Phi \geq \Phi_c$), critical ($0 \leq \Phi < \Phi_c$), collapse ($\Phi < 0$). The biological cardiac row uses per-record thresholds with the cluster center near $\Phi_c = 0$; the 100-class CNN entry within the computational row uses $\Phi_c = 0.082$. Geophysical is represented as one substrate family containing the three Donna Lea same-sensor points and the separate Tohoku foreshock-catalog point. The LLM row is shown in a separate observational-support marker style (substrate-shared structural-role supporting evidence) rather than under the regime color scheme, because LLM $\Phi$ values are not threshold-gated in this paper. Data derived from Tables 4 and 5 (Section 3).

# 4 The Observed $(\alpha, \Phi_c)$ Clustering

The per-substrate results reported in Section 3 use a common value of $\alpha = 0.1$ across engineered-cluster substrates, and a common reference threshold $\Phi_c = 0.25$ across six substrate families in the evidence: mechanical bearings, aerospace turbofans, electrical grids, geophysical monitoring (both the Donna Lea strainmeter and Tohoku foreshock catalog instantiations), quantum qubits, and computational neural-network architectures at standard class counts. Two specialized or substrate-specific threshold regimes appear within the remaining evidence. The 100-class CNN supervision case (a specialized regime within the computational substrate) uses $\Phi_c = 0.082$ as its operational threshold, with $\Phi_c = 0.084$ as the edge-case architecture's exact $\Phi$ value. The biological cardiac substrate uses per-record threshold calibration at a target false-positive rate of 0.01 on NORMAL windows rather than a fixed $\Phi_c$; calibrated per-record thresholds cluster near $\Phi = 0$.

The large-language-model substrate uses $\Phi_c = 0.25$ as a logged reference only, with no threshold-gated decision, and is not treated as a threshold-clustering data point for this section. This section reports the $(\alpha, \Phi_c)$ clustering observed in the tested substrates. The clustering is observed, not derived. What determines $(\alpha, \Phi_c)$ for a given system class is left as an open question; this paper does not propose a classification scheme for system types.

## 4.1 Engineered Steady-State Cluster

The engineered-cluster anchor is $\alpha = 0.1$ and $\Phi_c = 0.25$. The substrate instantiations reported in Section 3 that use these parameters without retuning are: mechanical bearings, aerospace turbofans, electrical grids, geophysical Donna Lea strainmeter cases, geophysical foreshock catalog, quantum qubits (both the 445-qubit snapshot and the 1,004-gate pair analysis), and computational neural-network architectures at standard class counts (2-class and 10-class supervision). Per the chronology reported in Section 2, $\alpha = 0.1$ and $\Phi_c = 0.25$ were first fixed in the origin repository `system-degradation-framework` at commit `5ec267f` dated 2025-10-21, and reused without modification in three subsequent production repositories through 2025-12-31 that validated engineered-cluster substrates; a later observational LLM repository reused both values in code on 2026-01-27, with the threshold logged only and not used to gate execution. A USPTO provisional patent application was filed on 2025-11-20 with source code from `system-degradation-framework` as the disclosed material, providing disclosure evidence independent of git history; the exact provisional-application identifier is reported in Section 11.

## 4.2 Specialized or Substrate-Specific Threshold Regimes

Two specialized or substrate-specific threshold regimes are documented in the consolidated evidence (the specialized computational 100-class CNN case and the biological cardiac case); a third preliminary regime (biological neural EEG) is documented later in this section. All three were documented at first validation; none is introduced later in the manuscript to fit subsequent cross-domain results.

The *specialized computational class* is the 100-class CNN supervision case in the `phi-controller` repository. Standard-class (2-class, 10-class) supervision uses $\Phi_c = 0.25$; the 100-class case uses $\Phi_c = 0.082$ operationally, with $\Phi_c = 0.084$ as the exact $\Phi$ of the edge-case viable architecture (Seed 83, 20.6% CIFAR-100 accuracy). The operational threshold of 0.082 was selected from a threshold-optimization sweep on 100-class training runs documented in the repository's canonical `experiments/OUTPUTS.md` at commit `77ab987` (2025-12-10); the shifted threshold was introduced at commit `a3e8ba2` (2025-12-10) and has been used unchanged since. The lower operational threshold is an empirical result of the threshold-optimization sweep documented in the repository; this paper does not derive the cause of the shift. $\alpha$ remains 0.1.

The *biological cardiac* regime in the `phi-bio-stability` repository uses $\alpha = 0.1$ but replaces the fixed $\Phi_c$ with per-record threshold calibration. For each patient record, a threshold is fit from NORMAL windows to achieve a target false-positive rate of 0.01. The calibrated per-record thresholds cluster near $\Phi = 0$; this near-zero clustering is a descriptive summary of where calibrated thresholds fall across records, not a single fixed threshold applied across records. The primary methodology statement is the per-record FPR-calibrated threshold. The substrate-specific threshold methodology was introduced at `phi-bio-stability` commit `a151789` (2026-02-02) and has been used unchanged since.

## 4.3 Preliminary Biological Neural Regime

A third regime, biological neural (EEG seizure-related signals), appears as preliminary evidence in the `phi-bio-stability` repository. Per the parameter chronology, EEG validation was introduced at commit `fa04607` (2026-02-04), which documented that cross-patient EEG validation required patient-specific calibration of both the coefficient and the sign direction of $\Phi$, with $\alpha$ near 0.55 in that exploratory work rather than the engineered-cluster value of 0.1. This EEG evidence is not included in the consolidated evidence table in Section 3 and is not claimed as a settled cross-substrate anchor; we report it here only to flag that an additional specialized regime appears to exist in biological neural data and has been documented at its first validation, not retuned. This EEG case is therefore treated only as preliminary calibration evidence, not as evidence for the shared engineered-cluster threshold. The paper does not treat the EEG $\alpha \approx 0.55$ as load-bearing evidence for a cross-domain clustering; it is mentioned to prevent the impression that the engineered cluster and the biological cardiac case exhaust the observed variation.

## 4.4 Observed, Not Derived

The $(\alpha, \Phi_c)$ pairs reported above cluster by substrate class in the tested evidence. This is an empirical observation on the validated substrates only. We do not propose a mechanism that determines $(\alpha, \Phi_c)$ from substrate properties, and we do not claim the clustering generalizes to substrates outside the tested sample. The presence of the 100-class CNN shifted threshold within the computational domain is worth noting: it means the shift from engineered-cluster parameters to shifted parameters is not a simple biology-versus-engineering dichotomy, since a specialized computational class also requires a shifted threshold. What determines $(\alpha, \Phi_c)$ for a given system class, and whether a deeper regularity underlies the observed clustering, is explicitly left as an open question for future work.

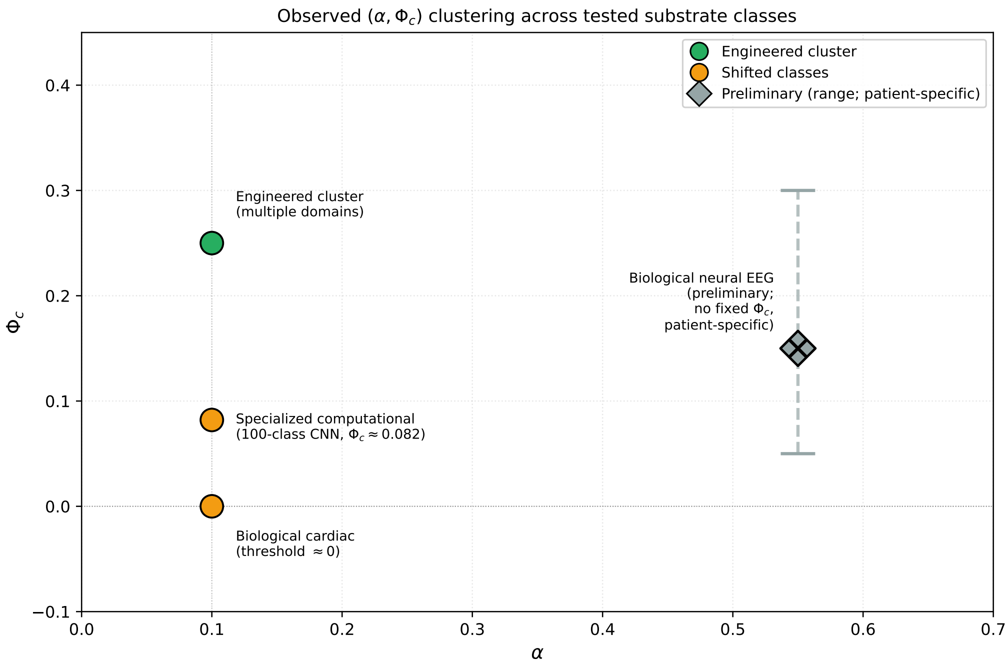


Figure 2: Observed $(\alpha, \Phi_c)$ clustering across tested substrates. Engineered-cluster substrates concentrate at $(0.1, 0.25)$; specialized computational (100-class CNN) appears at $(0.1, 0.082)$ operational; biological cardiac appears at $(0.1, \approx 0)$ where the threshold is per-record calibrated to FPR $= 0.01$ rather than fixed; biological neural (EEG) is shown separately as preliminary, non-load-bearing evidence with patient-specific calibration. The clustering is observed in the tested substrates only; what determines $(\alpha, \Phi_c)$ for a given system class is left as an open question in this paper. Data derived from the chronology in Table 2 (Section 2) and threshold conventions in Section 4.

# 5 Transition Detection Versus Stable-State Classification

The empirical results in Section 3 report $\Phi$ values across heterogeneous substrates in which stable and failing regimes are separated by the scalar. A reader might ask a narrower question: does $\Phi$ separate any two different operating states, or does it specifically separate transitions into a degraded regime from ongoing operation within some regime? This section reports the paired comparison that bounds this question. Two cardiac ECG tasks were evaluated under matched protocols on the same instrument family (RR-interval analysis from ECG annotation streams): an arrhythmia-detection task treated here as transition-positive under the RR-interval protocol, and an atrial-fibrillation-classification task treated here as a persistent alternative-rhythm label. The results differ sharply. We read this contrast as the paper's scope bound on $\Phi$: the scalar reports the transition-positive task under this protocol, not the persistent alternative-rhythm classification task.

## 5.1 Matched Protocol

Both tests use the same $\Phi$ coordinates ($I$ = baseline_std_RR/current_std_RR, $\rho$ = autocorrelation at lag 1 of the RR-interval series, $S$ = Shannon entropy in bits of the

RR-interval distribution), the same $\alpha = 0.1$, the same 60-second window length, and the same per-record threshold calibration methodology (threshold fit from NORMAL windows to a target false-positive rate of 0.01). The principal differences are the dataset, annotation policy, and target label. The arrhythmia test (MIT-BIH Arrhythmia Database, PhysioNet) labels windows as positive when they contain annotated arrhythmia beats (symbols A, E, F, J, Q, S, V, a); normal windows contain only sinus-rhythm beats (symbols L, N, R, e, j). The atrial-fibrillation test (MIT-BIH Atrial Fibrillation Database, PhysioNet) labels windows as positive when they are annotated as AFIB under a strict rhythm policy; normal windows are annotated as non-AFIB rhythm. The canonical artifacts are, for arrhythmia, `results/test1_mitdb_cardiac/mitdb_phi_summary.json` at commit `a440b76` (2026-02-07) in the `phi-bio-stability` repository, and for AFib, `results/test2_afdb_afib/afdb_phi_summary.json` at commit `5daba22` (2026-02-03) in the same repository. Both summary-JSON audit records indicate the same ECG-based $\Phi$ pipeline structure, with shared coordinate definitions, $\alpha$, window length, and per-record FPR-based calibration methodology. The arrhythmia reference implementation is `scripts/test_phi_ecg_mitdb.py` at commit `a151789`; the AFib reference implementation is `scripts/test_phi_ecg_afdb_rhythm.py` at commit `5daba22`.

## 5.2 Positive Case: Arrhythmia Transition Detection

The MIT-BIH Arrhythmia Database processed 34 records out of 48 total (14 records skipped because they lacked pure NORMAL windows needed for per-record threshold calibration; these are records with severe or continuous arrhythmias). Across 1,022 sixty-second windows (461 arrhythmia, 561 normal), $\Phi$ achieved AUC = 0.9148, precision = 0.9171, recall = 0.7202, specificity = 0.9465, F1 = 0.8068, and MCC = 0.6936. The confusion matrix at the per-record calibrated threshold is TP = 332, FP = 30, TN = 531, FN = 129. The per-record thresholds calibrated to FPR = 0.01 on NORMAL windows cluster near $\Phi = 0$; this is the sense in which the cardiac threshold regime is summarized as clustering near $\Phi_c \approx 0$ in Section 4. The primary method statement throughout the paper is the per-record FPR-calibrated threshold; the near-zero clustering is a descriptive summary of where those thresholds fall.

## 5.3 Negative Case: Atrial Fibrillation Classification

The MIT-BIH Atrial Fibrillation Database processed 21 records out of 25 requested (2 failed on sampto/sampfrom errors during data retrieval; 2 were skipped due to no pure NORMAL windows). Across 11,988 sixty-second windows (4,071 AFib, 7,917 normal), $\Phi$ achieved AUC = 0.5556, precision = 0.6602, recall = 0.0420, specificity = 0.9889, F1 = 0.0790, and MCC = 0.1006. The confusion matrix is TP = 171, FP = 88, TN = 7,829, FN = 3,900. The AUC of 0.5556 is close to chance, and the recall of 0.0420 indicates the calibrated threshold rarely classifies AFib windows as positive. $\Phi$ does not separate AFib from non-AFIB rhythm in this test.

## 5.4 The Scope Bound

The two tests share the instrument family, the coordinate definitions, $\alpha$, the window length, and the per-record threshold calibration methodology; their principal differences are the dataset, annotation policy, and target label. On the arrhythmia task, $\Phi$ produces

AUC = 0.9148 and MCC = 0.6936. On the AFib task, $\Phi$ produces AUC = 0.5556 and MCC = 0.1006. We read this asymmetry as a scope statement. Arrhythmia windows, as labeled in MIT-BIH, contain beat-level rhythm disturbances relative to the normal-symbol windows used for calibration; under this protocol, those windows function as annotation-defined transition events for the RR-interval scalar. AFib, by contrast, is treated here as a persistent alternative-rhythm label under the strict rhythm policy: within an AFib-labeled window, the protocol tests rhythm-state classification rather than detection of departure from a calibrated normal baseline. The scalar $\Phi$ in its tested form reports the former and not the latter. This result, on one matched positive-negative pair in one biological dataset, is the paper's sharpest scope-bounding observation.

Two qualifications are important. First, the scope bound is demonstrated by one paired comparison on biological data; we do not claim this generalizes into a substrate-independent distinction between transitions and alternative modes. Second, the AFib recall of 0.0420 means $\Phi$ rarely classifies AFib windows as positive under the calibrated threshold; this is not a statement that $\Phi$ *cannot* separate AFib from non-AFIB rhythm under any coordinate definition or calibration, only that $\Phi$ under the matched protocol used here does not. Future work that redefines $I$, $\rho$, or $S$ specifically for rhythm-classification would be a separate study.

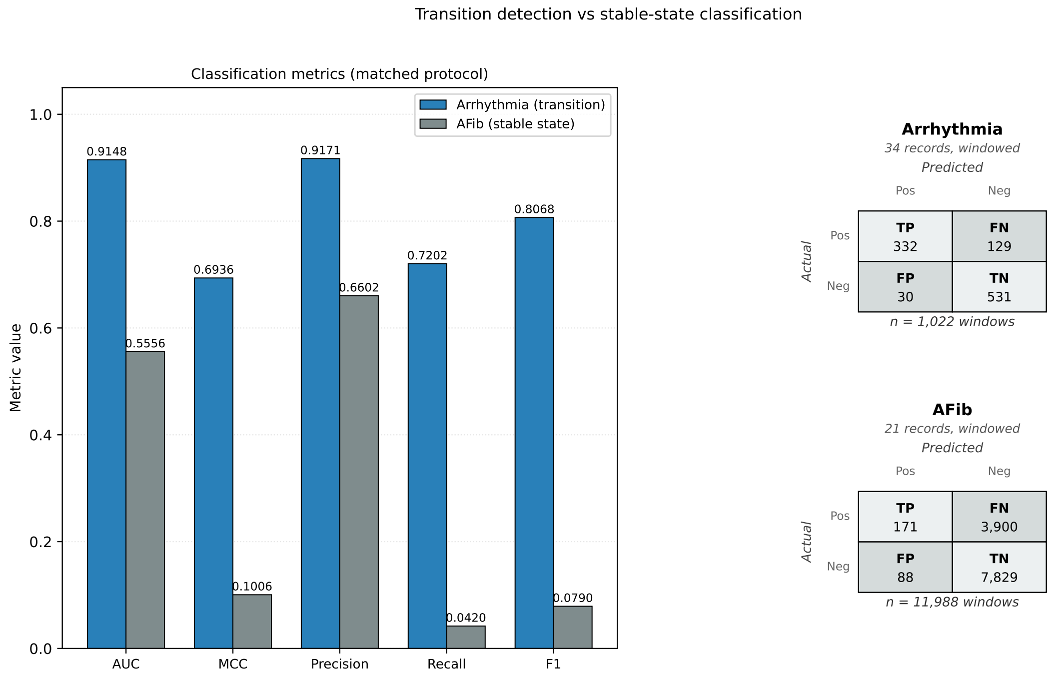


Figure 3: Transition detection versus stable-state classification under matched protocol. Left: arrhythmia detection (MIT-BIH Arrhythmia, 34 records processed, 1,022 windows). $\Phi$ separates arrhythmia transition-positive windows from normal-symbol windows with AUC = 0.9148, MCC = 0.6936. Right: atrial fibrillation classification (MIT-BIH Atrial Fibrillation, 21 records processed, 11,988 windows). $\Phi$ does not separate AFib from non-AFIB rhythm (AUC = 0.5556, MCC = 0.1006, recall = 0.0420). Both tests use the same RR-interval-based coordinates, the same $\alpha = 0.1$, the same 60-second window length, and the same per-record threshold calibration at FPR = 0.01. The asymmetry bounds the scope of $\Phi$ in its tested form: annotation-defined transition detection under this protocol, not classification between persistent alternative-rhythm labels. Data derived from the arrhythmia and AFib summary artifacts cited in Section 5.

# 6 Operational Recurrence Across Implementations

The cross-domain empirical results reported in Section 3 establish regime-separation evidence for the mechanical, aerospace, electrical, geophysical, quantum, and computational substrate families, scope-boundary evidence for the biological cardiac substrate, and observational structural-role evidence for the large-language-model substrate. Independently of those results, the same scalar has been used in four operational modes across the evidence and operational-recurrence repositories. The three audited operational-recurrence repositories (`phi-hybrid-allocation`, `phi-controller-quantum`, `phi-objective-controller`) reuse $\alpha = 0.1$ and $\Phi_c = 0.25$ unchanged from the engineered-cluster anchor; the detection mode additionally includes the specialized or substrate-specific threshold regimes already documented in Section 4 (100-class CNN at $\Phi_c = 0.082$; biological cardiac with per-record threshold calibration). This section reports the operational recurrence. It is portfolio-level observation, not independent evidence for the scalar itself; the operational-recurrence observation is explicitly non-load-bearing relative to the regime-separation and scope-boundary evidence.

## 6.1 Detection

The first operational mode is threshold-based regime classification, which is the mode reported throughout Section 3. The scalar is computed from substrate-appropriate observables and compared against $\Phi_c = 0.25$ (or the specialized/substrate-specific threshold per Section 4) to assign a stable, critical, or collapse regime label. This mode is implemented in `-thermodynamic-stability-prediction` for mechanical, aerospace, electrical, and geophysical substrates; in `quantum-phi-validation` for quantum qubits; in `phi-controller` for computational neural-network architectures; and in `phi-bio-stability` for biological cardiac records. Detection is the baseline operational mode on which the other three modes build.

## 6.2 Routing

The second operational mode uses $\Phi$ to route computation at execution time. In the `phi-hybrid-allocation` repository, $\min(\Phi)$ across a circuit's qubits is used to route quantum computations between quantum hardware (when $\min(\Phi) \geq \Phi_c$) and classical simulation (when $\min(\Phi) < \Phi_c$). Under a strict validation protocol with verified physical-qubit binding and decisive ground-truth checks, the routing mode was evaluated on seven quantum algorithms on IBM hardware: HIGH-$\Phi$ vs LOW-$\Phi$ error ratios range from 2.90× (quantum phase estimation) to 30.47× (Bernstein-Vazirani), with intermediate values 4.07× (Simon's), 7.38× (GHZ), 8.12× (quantum Fourier transform), 12.97× (Deutsch-Jozsa), and 15.90× (Grover). Canonical aggregation is documented in `Outputs_MD/strict/README.md` in the `phi-hybrid-allocation` repository; individual test scripts for the seven algorithms are in `experiments/strict/`. Parameters are $\alpha = 0.1$ and $\Phi_c = 0.25$ reused unchanged.

## 6.3 Control (Quantum)

The third operational mode uses $\Phi$ to select qubits prior to quantum circuit execution as an intervention against degradation. In the `phi-controller-quantum` repository, a baseline test on LOW-$\Phi$ qubits on `ibm_fez` (qubits 149 and 150 with $\Phi$ values 0.097 and 0.030) produced an error rate of 18.36%. A controller-selected HIGH-$\Phi$ pair on the same backend (qubits 138 and 151 with $\Phi$ values 0.9988 and 0.9984) produced an error rate of 2.73%, a relative error reduction of 85.1%. The same repository's Test 8 reports a 68.7% relative reduction of $\Phi$-based selection compared to selection by raw $T_2$ only on the same backend, and a cross-backend test across `ibm_fez`, `ibm_torino`, and `ibm_marrakesh` documents $\Phi$-based selection winning on two of three backends with an aggregate net of +6.37 percentage points in $\Phi$'s favor; the `ibm_marrakesh` counterexample is acknowledged in Section 9. Parameters are $\alpha = 0.1$ and $\Phi_c = 0.25$ reused unchanged.

## 6.4 Control (General)

The fourth operational mode uses $\Phi$ as an objective for action selection during system operation. In the `phi-objective-controller` repository, a trajectory-aware controller selects among available actions (for neural training: learning-rate adjustments and no-op;

for quantum: qubit migration; for bearings: speed reduction and lubrication) to improve expected $\Phi$ trajectory subject to task-performance constraints. On fully out-of-sample two-fold cross-validation of neural training (CIFAR-10, ResNet-18 (5), 20 seeds, 192 training rows per fold), the controller achieved 12/20 wins (60%) with +0.073% mean delta relative to baseline; the worst observed loss was −0.17%, substantially smaller than the largest gain of +0.49%. The same controller architecture was also instantiated in the same repository on IBM quantum hardware in a 9-trial milestone run (3 seeds across 3 backends: 8 wins, 1 loss, 0 ties; mean delta +4.69%). It was also instantiated on a physics-based bearing simulator, where the surrogate controller won 10 of 10 simulated intervention trials with +0.067 mean $\Phi$ delta; this simulated setting was used because the XJTU-SY bearing data are run-to-failure observations with no real intervention actions. These results are reported as architecture-transfer recurrence rather than as independent regime-separation evidence. Parameters are $\alpha = 0.1$ and $\Phi_c = 0.25$ reused unchanged across the Bearings, Quantum, and Neural modules in the repository. Transfer-test caveats on this controller architecture are documented in Section 9.

### 6.5 What the Recurrence Supports

The four operational modes above use the same scalar $\Phi = I \cdot \rho - \alpha \cdot S$. The three audited operational-recurrence repositories (`phi-hybrid-allocation`, `phi-controller-quantum`, `phi-objective-controller`) retain $\alpha = 0.1$ and $\Phi_c = 0.25$ unchanged from the engineered-cluster anchor; the detection mode draws on repositories whose threshold conventions (engineered-cluster and specialized or substrate-specific) are documented in Section 4. The modes are operationally distinct: detection is threshold classification on a computed regime score for the analyzed system or window, routing is selection among execution alternatives using $\min(\Phi)$, control-quantum is pre-execution intervention against degradation, control-general is action selection during operation under task-performance constraints. The recurrence supports the observation that the scalar has been reused across these operationally distinct implementations without parameter retuning in the audited operational-recurrence repositories. It does not establish that the scalar is optimal for any of these operational modes, that it generalizes beyond the tested substrates within each mode, or that the operational modes are exhaustive of the scalar's possible uses. The operational-recurrence observation is reported here for completeness of the portfolio-level picture; the paper's load-bearing scientific results (regime separation and the transition-versus-stable-state scope bound) are established independently in Sections 3 and 5.

## 7 Forward Prediction and Cold Test Result

The cross-domain results in Section 3 are retrospective. A natural concern is that retrospective evaluations can absorb implicit tuning, even when the scalar coordinates and parameters are formally locked in prior code. To address this, a single preregistered forward test was executed before the cold-test result was incorporated into this manuscript. The test was preregistered in a committed artifact whose commit timestamp predates the computation; the execution script was committed to a tracked repository before it was run; and the success criterion was locked in advance.

## 7.1 Preregistered Protocol

The selected system was XJTU-SY Bearing2_1, operating condition 37.5 Hz / 11 kN, 491 vibration CSV files from the published XJTU-SY Bearing Dataset (24). Bearing2_1 was not among the 10 previously validated XJTU-SY bearings in the `-thermodynamic-stability-prediction` bearing pipeline; it shares operating condition with four of those bearings (Bearing2_2, 2_3, 2_4, 2_5), so the existing bearing pipeline already covered that operating condition and no protocol adaptation was made for the new bearing.

The observable mappings were locked verbatim from the canonical bearing pipeline: $I$ = baseline_RMS / current_RMS, $\rho$ as the autocorrelation of the RMS time series at lag 1, and $S$ as the Shannon entropy in bits of the RMS signal distribution. The parameters were locked at $\alpha = 0.1$ and $\Phi_c = 0.25$, inherited from the engineered-cluster anchor documented in Section 2. The success criterion was stated in a single sentence before the test was run: the end-of-life $\Phi$ value for Bearing2_1 will fall below 0.25. The failure criterion was the complementary condition, $\Phi \geq 0.25$. Two ambiguity escape hatches were disclosed in advance in the preregistration: a no-signal outcome (no degradation structure detected in the RMS series) and a protocol-breaking outcome (numerical issues in the computation). Neither was exercised in this test.

## 7.2 Transparent Disclosure of Prior Signal Exposure

Bearing2_1 had previously been analyzed in the `system-degradation-framework` repository under a mathematically distinct adaptive-threshold computation on the same raw vibration signal. That earlier computation produced an adaptive-threshold modulator used for detection timing, not the scalar $\Phi$ reported in this paper, and it did not determine the value of $\Phi = I \cdot \rho - \alpha \cdot S$ on this bearing. The present result is therefore the first preregistered evaluation of the scalar $\Phi$ on Bearing2_1, not a first exposure of the underlying vibration signal to portfolio analysis; we describe it as a preregistered forward test of the scalar rather than as a blind test, and the evidentiary value lies in the preregistered commitment chain (preregistration committed before script, script committed before execution) rather than in the prediction's difficulty. The preregistration document discloses this prior exposure explicitly and distinguishes the two computations.

## 7.3 Commitment Chain and Execution

The provenance chain is a sequence of committed artifacts in git history:

- Preregistration document `COLD_TEST_PREREGISTRATION.md`, committed at hash `427f2d7` on 2026-04-19 in the working repository before any cold-test computation.
- Cold-test script `cold_test_bearing2_1.py`, committed at hash `2a2bf3f` on 2026-04-20 before execution. The script copied the scalar computation functions verbatim from the canonical bearing pipeline's `paper_validation/scripts/compute_bearing_components.py` in the `-thermodynamic-stability-prediction` repository at source commit `2e2099c`; the source commit hash is embedded in both the script header and the result JSON. The standalone-script execution path was used because the canonical file contains top-level side effects on import that would affect other tests in the bearing pipeline; the scalar functions themselves were copied verbatim from the canonical file.

- Result JSON `cold_test_bearing2_1_result.json`, committed at hash `c521ae6` on 2026-04-20 with the execution output.
- Result artifact `COLD_TEST_RESULT.md`, committed at hash `5da5322` on 2026-04-20 reporting the verdict.
- Amendment document `COLD_TEST_AMENDMENT.md`, committed 2026-04-22, documenting five post-hoc clarifications (execution path used a standalone script rather than a modification of the canonical file; the verdict logic was binary rather than exercising the four preregistered outcome classes because the outcome was unambiguous; a full $\Phi$ trajectory was not produced as a separate artifact because the preregistered threshold gate is on the end-of-life value only; chronology wording was harmonized with Section 2; and the preregistration commit hash placeholder was resolved). None of the amendments affected the mathematics, data source, computation functions, or the locked parameters.

## 7.4 Result

The end-of-life $\Phi$ value for Bearing2_1 is $-0.342137$, satisfying the preregistered criterion $\Phi < 0.25$.

The component values at end-of-life are $I = 0.073596$, $\rho = 0.995781$, and $S = 4.154230$. The descriptive diagnostic of degradation structure is the baseline-to-final RMS ratio: the final RMS (7.161) is 13.588 times the baseline mean RMS (0.527), confirming that substantial degradation signal was present in the time series rather than a no-signal condition. Bearing2_1's end-of-life $\Phi = -0.342$ sits near the end-of-life $\Phi = -0.370$ of Bearing2_3, which shares the 37.5 Hz / 11 kN operating condition; both sit in the collapse regime ($\Phi < 0$), consistent with the end-of-life $\Phi$ range $-0.370$ to $-0.003$ observed across the 10 previously validated XJTU-SY bearings in Section 3.

## 7.5 What This Result Does and Does Not Show

The preregistered prediction that $\Phi < 0.25$ is a coarse preregistered threshold prediction in the following sense: a run-to-failure XJTU-SY bearing, evaluated by the scalar formula on its end-of-life vibration window, is expected to produce small or negative $\Phi$ by the structure of the formula itself, because large RMS growth drives $I$ toward zero and entropy-driven subtraction drives $\Phi$ further downward. The evidentiary value of the cold test is not that the prediction was difficult. The evidentiary value is in the preregistered commitment chain: the prediction was committed before the scalar cold-test output reported here was computed; the execution script was committed before it was run; the parameters were locked to values fixed in a different production repository more than five months before this cold test (Section 2); and the outcome was binary against a single-sentence preregistered criterion. These ordering constraints are documented by git history rather than left to the authors' good faith.

The cold test does not demonstrate that the scalar will generalize to every future bearing, every future substrate, or every future failure mode. It demonstrates that on one concrete preregistered test, with the preregistration and execution-script commits recorded before the reported computation, the scalar's end-of-life prediction held. This single cold test is one data point, reported with the prior-exposure disclosure and commitment-chain qualifications above.

# 8 Discussion

This paper reports an empirical regularity: the scalar $\Phi = I \cdot \rho - \alpha \cdot S$, with $\alpha = 0.1$ and $\Phi_c = 0.25$ for engineered-cluster substrates, recurs across the tested substrates in three distinct forms of evidence. First, across the mechanical, aerospace, electrical, geophysical, quantum, and computational substrate families, $\Phi$ separates operationally stable regimes from failing or critical regimes on documented failure outcomes. Second, biological cardiac evidence provides a scope-boundary result: under matched protocol, $\Phi$ separates arrhythmia-transition windows from normal-symbol windows but does not separate atrial fibrillation from non-AFIB rhythm. Third, large-language-model evidence provides observational structural-role support under matched prompt suites, without threshold-gated regime decisions. A preregistered forward test on a mechanical system not in the original validation set satisfied the preregistered end-of-life threshold criterion. The paper reports the pattern; it does not claim to explain it.

## 8.1 What the Cross-Substrate Pattern Implies for Practice

Cross-domain monitoring currently uses substrate-specific tools: vibration-based indicators for bearings, engine-health models for turbofans, grid-frequency detectors for power systems, change-point detection for time-series anomalies, calibration-based quality scores for qubits, early-stopping heuristics for neural training, RR-interval classifiers for cardiac monitoring, and perplexity-based or embedding-based drift detectors for language models. These tools are built and tuned independently by different communities. The empirical result reported here is that a single three-coordinate scalar, with one pair of parameters for engineered-cluster substrates and substrate-specific instantiations of the coordinates, tracks stable-versus-failing regime distinctions across the mechanical, aerospace, electrical, geophysical, quantum, and computational substrate families, tracks arrhythmia transition-positive windows but not the persistent alternative-rhythm AFib classification task in the biological cardiac substrate per Section 5, and produces paired observational contrasts across prompt conditions in the LLM substrate. The practical implication is modest: a unified scalar is available as an additional monitoring signal that can be computed from the same observables used by existing substrate-specific tools, with parameters locked in advance per Section 2. We do not claim the scalar replaces substrate-specific tools; we claim it recurs as a consistent additional signal across the tested substrates within the scope bounds reported in Sections 3, 4, and 5.

## 8.2 Chronology and the Reused-Unchanged Discipline

Section 2 documents that $\alpha = 0.1$ and $\Phi_c = 0.25$ were first fixed in production code on 2025-10-21 and reused unchanged in three subsequent production repositories through 2025-12-31; a later observational LLM repository reused both values in code on 2026-01-27, with the threshold logged only and not used to gate execution. A USPTO provisional patent application filed on 2025-11-20 provides disclosure evidence independent of git history. Specialized or substrate-specific threshold regimes ($\Phi_c = 0.082$ operational for 100-class CNN supervision; per-record threshold calibration at FPR $= 0.01$ for biological cardiac; $\alpha \approx 0.55$ for preliminary biological neural EEG) appear in git history together with the evidence that motivated them, so these regimes were documented at first validation rather than retuned after the fact. The code-to-paper audit verified this chronology across the evidence repositories and surfaced no reuse inconsistencies. Collectively, this addresses the concern that the cross-domain empirical results reflect implicit per-domain tuning.

## 8.3 The Transition Scope Bound

Section 5 reports the paper's sharpest scope-bounding result: under matched RR-interval-based protocols with $\alpha = 0.1$ and per-record FPR-calibrated thresholds, $\Phi$ separates arrhythmia-transition windows from normal-symbol windows (AUC = 0.9148, MCC = 0.6936) but does not separate atrial fibrillation from non-AFIB rhythm (AUC = 0.5556, MCC = 0.1006). We read the asymmetry as a scope statement on what $\Phi$ reports under this protocol: annotation-defined transition detection, not classification between persistent alternative-rhythm labels. This is narrower than "$\Phi$ detects all cardiac anomalies," which would be false on this evidence. It is also sharper than "$\Phi$ detects failures," which leaves undefined what counts as a failure. The paper reports transition detection under the matched protocol as the demonstrated scope.

## 8.4 The Preregistered Forward Test

Section 7 reports a preregistered forward test on XJTU-SY Bearing2_1, which was not in the original 10-bearing validation set. The end-of-life $\Phi$ value was $-0.342137$, satisfying the preregistered criterion $\Phi < 0.25$. The evidentiary value of this test is not the difficulty of the prediction, which was coarse; it is the preregistered commitment chain documented by git history (preregistration committed before script, script committed before execution, parameters locked to values fixed in a different repository months earlier). We disclose in that section that the same raw vibration signal had prior portfolio exposure under a mathematically distinct adaptive-threshold computation; the present result is the first preregistered evaluation of the scalar $\Phi$ on this bearing, not a first exposure of the underlying vibration data.

## 8.5 Operational Recurrence as Non-Load-Bearing Context

Section 6 reports that the same scalar has been reused across four operational modes across the evidence and operational-recurrence repositories. The three audited operational-recurrence repositories retain $\alpha = 0.1$ and $\Phi_c = 0.25$ unchanged; the detection mode includes the specialized or substrate-specific threshold regimes already documented in Section 4. The recurrence is portfolio-level context, not independent scientific evidence for the scalar itself. The paper's load-bearing results rely on the cross-domain regime separation and the transition scope bound, not on the operational recurrence.

## 8.6 What This Work Does Not Claim

The paper reports a pattern observed across the tested substrates. It does not claim that the pattern generalizes to substrates outside the tested sample. It does not propose that the $(\alpha, \Phi_c)$ clustering reported in Section 4 is a classification scheme for system types; the clustering is observed, not derived. It does not assert that the scalar identifies a conserved quantity or obeys a conservation law. It does not claim the scalar is optimal for any specific monitoring task or that it outperforms substrate-specific tools in cases where those tools are already mature. The open questions stated in Section 4 about what determines $(\alpha, \Phi_c)$ for a given system class, and whether a deeper regularity underlies the observed clustering, are explicitly left for future work. The paper is designed as a cross-domain empirical regularity report, not as a theoretical contribution.

# 9 Limitations

This section documents the scope boundaries and failure modes of the evidence reported in this paper. Where a failure mode is visible in the code artifacts cited elsewhere in the paper, it is disclosed here rather than buried.

## 9.1 Scope Limitations

The cross-domain pattern reported in Section 3 is established on the eight tested substrates. It is not tested on substrates outside this sample. The per-domain sample sizes are substrate-appropriate but small in absolute terms (10 bearings, 10 turbofans, 2 grids, 3 same-sensor geophysical events plus one foreshock catalog, 445 qubits, 660 neural-network architectures in the canonical `phi-controller` validation summary, 34 processed patient records of 48 total in MIT-BIH Arrhythmia, and 8 LLM test configurations on 3 models). Stable controls are limited in several domains: the XJTU-SY bearing dataset and the NASA C-MAPSS turbofan dataset are run-to-failure by construction with no separate stable-system cohort, and cross-domain stable-versus-failing paired comparisons are available in the electrical grid and geophysical Donna Lea substrates but not in the bearing and turbofan substrates. Most of the evidence is retrospective (failures were documented in the datasets before $\Phi$ was computed); the single exception is the preregistered forward test on Bearing2_1 reported in Section 7. The transition-versus-stable-state scope bound in Section 5 is demonstrated by one matched positive-negative pair on biological cardiac data; we do not claim it generalizes to other substrates or other instrument families.

## 9.2 Documented Failure Modes and Counterexamples

Several failure modes and counterexamples appear in the code artifacts cited by this paper. Each is disclosed here explicitly.

**Bearing1_4 near-threshold case.** In the bearing validation reported in Section 3, Bearing1_4's end-of-life $\Phi$ value is $-0.003$, essentially at the collapse-critical boundary. This bearing is a run-to-failure case, so the label is correct, but a threshold-based classifier at $\Phi_c = 0$ would classify this case inconsistently across small parameter perturbations.

**CIFAR-10 CNN false kills.** In the computational neural-network evidence reported in Section 3, the CIFAR-10 CNN test in the `phi-controller` repository produced two false kills (architectures at 60.6% and 61.5% final accuracy were killed at $\Phi_c = 0.25$, while the threshold was intended for architectures below 60%). Kill precision on that test was 98.0%.

**Quantum `ibm_marrakesh` same-day counterexample.** In the quantum evidence reported in Section 3, a single-day evaluation on `ibm_marrakesh` produced a pair selected by $\Phi$ that yielded higher error than a pair selected by raw $T_2$ only. Across three backends on the same day (`ibm_fez`, `ibm_torino`, `ibm_marrakesh`), $\Phi$-based selection won on two of three with an aggregate net of +6.37 percentage points in $\Phi$'s favor; `ibm_marrakesh` was the counterexample. This counterexample is disclosed here and does not support an assertion that $\Phi$-based selection dominates raw $T_2$ in every instance.

**Quantum Bell-state and variational-circuit mixed results.** In the pinned `quantum-phi-validation` artifacts, the Bell-state test was inconclusive and opposite to the single-qubit $\Phi$ prediction: LOW-$\Phi$ pairs had mean error 4.92% while HIGH-$\Phi$ pairs had mean error 7.06%, with the artifact attributing the inversion to a bad two-qubit gate in one HIGH-$\Phi$ pair. The variational-circuit test was also weak or mixed: TVD consistency inverted at several ansatz depths, while entropy showed only a small effect in the expected direction. These cases suggest that single-qubit $\Phi$ does not fully capture two-qubit gate quality or entangling-circuit behavior. The two-qubit gate-error analysis cited in Section 3 relies on calibration-based gate-error data rather than on a claim that single-qubit $\Phi$ dominates all circuit-level outcomes.

**AFib classification near-chance result.** In Section 5, the atrial fibrillation classification test produced AUC = 0.5556 and recall = 0.0420, meaning $\Phi$ under the matched protocol rarely classifies AFib windows as positive under the calibrated threshold. This is reported as the paper's scope bound, not as a failure the paper seeks to fix.

**Controller transfer-test caveats.** In Section 6, the `phi-objective-controller` cross-validation result (12/20 wins at +0.073% mean delta) applies to fully out-of-sample cross-validation folds within the original seed 1–20 population. An out-of-range transfer test on seeds 21–40, with the same surrogate architecture trained on seeds 1–20, produced 9/20 wins at −0.019% mean delta (mixed, near-neutral). An Adam-optimizer transfer pilot on 5 seeds produced 2/5 wins at +0.044% mean delta (preliminary only). These transfer results are reported here as transfer-test caveats.

**EEG patient-specific calibration requirement.** The preliminary biological neural evidence mentioned in Section 4 requires patient-specific calibration of both the coefficient $\alpha \approx 0.55$ and the sign direction of $\Phi$; the EEG result is not claimed as a training-free cross-patient cross-domain anchor and is not included in the consolidated evidence table.

## 9.3 Methodology Qualifications

Cross-domain recurrence is reported across heterogeneous unit types per the unit-of-analysis rule in Section 2. Per-domain results are not pooled into a single statistical sample. The substrate-shared structural-role observation is demonstrated on the tested implementations and does not assert that arbitrary implementations of $I$, $\rho$, and $S$ in untested substrates will produce a regime-separating $\Phi$; implementations must satisfy the admissibility rules in Section 2. The biological cardiac threshold regime uses per-record FPR-calibrated thresholds rather than a fixed $\Phi_c = 0.25$, which differs methodologically from the engineered-cluster threshold convention used in the other load-bearing substrates. The LLM substrate is observational; no $\Phi$ threshold gates execution in those tests, and LLM evidence supports the substrate-shared structural-role observation rather than the regime-separation evidence.

The preregistered Bearing2_1 cold test reported in Section 7 should be read with the prior-exposure disclosure made there: the bearing had appeared previously under a mathematically distinct adaptive-threshold computation in `system-degradation-framework`. The evidentiary value of the test is therefore the preregistered commitment chain documented by git history rather than a claim that the dataset had never appeared in earlier work.

# 10 Related Work

This paper sits adjacent to several research traditions. Each overlaps with part of the present study, but none addresses the same combination of fixed scalar form, pinned-commit provenance, and cross-substrate recurrence reported here. This section names the nearest neighbors and marks the distinctions.

## 10.1 Critical Transitions and Early-Warning Signals

The oldest tradition adjacent to this work is the critical-transitions literature from ecology and climate science (19; 20; 4). That literature observes that systems approaching bifurcations exhibit generic statistical signatures (critical slowing, rising variance, increased autocorrelation) that can be measured from time-series data without knowing the underlying dynamical equations. The coordinate $\rho$ in this paper's scalar is conceptually adjacent to the critical-slowing-down signal: both quantify temporal persistence. The difference is scope and form. The critical-transitions literature establishes statistical signatures that appear near specific classes of bifurcation in specific substrate families (ecosystems, climate subsystems, population dynamics). This paper reports a fixed three-coordinate scalar, with parameters locked before cross-domain validation, tracked across the tested substrate families, which are not uniformly described by bifurcation theory. The two efforts share a concern with generic signals of approaching transition but do not share a formal framework.

## 10.2 Change-Point Detection

Change-point detection (16; 1; 21) provides methods for locating times at which the statistical properties of a time series shift. These methods are domain-general in the sense that they operate on any real-valued sequence, but they report a change location rather than a regime classification, and they require a distribution-shift hypothesis appropriate to the application. The scalar $\Phi$ reported here is a regime indicator, not a change-point detector: it produces a value at each observation window that can be compared to a threshold to classify regime, rather than identifying the moment when regime changes. Change-point detection and regime classification are complementary rather than competing tools.

## 10.3 Anomaly Detection

Anomaly detection in machine learning (2; 17) produces a score for each observation indicating how far it deviates from a learned distribution of normal examples. This is closer in spirit to $\Phi$ as a regime indicator, but the two differ methodologically. Anomaly detectors are typically trained on data. The scalar $\Phi$ is not trained; its coordinates are computed directly from observables under the admissibility rules of Section 2, and the threshold $\Phi_c = 0.25$ is a fixed value from the engineered-cluster anchor, not a learned boundary. Anomaly detection is therefore a broader class of methods with a training-data dependency; the scalar reported here is a specific training-free instance of the regime-indicator pattern.

## 10.4 Health Monitoring and Prognostics

Health-monitoring and prognostics research in mechanical and aerospace engineering uses domain-specific feature engineering (vibration spectra, RUL estimation, degradation trajectory fitting) to predict remaining useful life or time to failure (8; 11). These methods are mature within their substrates. The distinction from the present work is scope: substrate-specific monitoring tools are built and tuned by substrate-specific communities, while the scalar reported here uses a substrate-shared structural form with substrate-specific coordinate instantiations. The scalar does not replace domain-specific monitoring; it tracks the same regime-separation structure using a unified coordinate form across substrates.

## 10.5 Prior Single-Substrate Stability Methods

Prior single-substrate stability methods have reported substrate-specific monitoring approaches, including seismic-monitoring and metal-fatigue prediction systems. These operate within one substrate with substrate-specific tuning. The present work differs in that the reported scalar is evaluated across the tested substrates using a shared coordinate-role structure with parameters locked before cross-domain validation, rather than within one substrate with substrate-specific tuning. Source identifiers for adjacent prior methods are listed in Section 11.

## 10.6 Prior Qubit Selection Work

Mapomatic (15) is a noise-aware qubit-selection and post-compilation mapping tool developed at IBM Quantum. It scores candidate circuit layouts using heuristic cost functions derived from device calibration data and reports recovery of approximately 40% of missing fidelity in benchmark circuits through improved qubit selection. The quantum evidence reported in Section 3 differs in scope: it includes the $T_2/T_1$ coherence-time ratio explicitly as the $\rho$ coordinate, with a Pearson correlation of $r = 0.9458$ between $\Phi$ and the published $T_2/T_1$ ratio across 445 qubits on three IBM backends, and evaluates $\Phi$ as a regime-classification signal rather than as a post-compilation routing tool. The scalar is therefore adjacent to Mapomatic-style calibration-aware routing, not a replacement for it.

## 10.7 Prior NAS Viability Work

Neural-architecture-search supervision has a long tradition of early-stopping heuristics (18) and adaptive-budget methods (12; 7). These methods compare model performance at fixed evaluation points against sliding-window criteria, allocating computational budget to architectures that appear more promising. The computational evidence reported in Section 3 uses $\Phi$ as a trajectory-aware regime classifier rather than as a budget-allocation rule: architectures are classified as viable, critical, or non-viable from the full training trajectory using the same coordinate structure as other substrates. The scalar is bounded by the documented false-kill cases in Section 9; it is not claimed to outperform Hyperband or Successive Halving in their respective budget-allocation roles.

### 10.8 Prior LLM Drift Detection

Recent research on large-language-model drift detection includes input-embedding shift analysis, token-level log-probability distribution tracking, and hidden-state-based drift indicators that require white-box access to internal model representations. The LLM evidence reported in Section 3 is observational and supports the substrate-shared structural-role observation rather than regime-separation evidence; the LLM tests evaluate cross-architecture paired comparisons under matched prompt suites on output embeddings without requiring internal model access. The scope is different: drift-detection methods are built for LLM drift as a detection task, while the scalar reported here is evaluated on LLMs as a consistency check of the coordinate-role structure across one additional substrate, not as a competing LLM-drift-detection method.

### 10.9 Differentiating Contribution

Much of the prior work above either operates within a single substrate or requires substrate-specific training, white-box access, or domain-tuned parameters. The present work reports a single scalar with a fixed coordinate-role structure and parameters locked before cross-domain validation, tracked across the tested substrate families using code-output artifacts at pinned commit hashes. The contribution is the cross-substrate recurrence of the scalar and the scope bounds documented in Sections 4, 5, and 9, not superiority on any individual substrate's native task. Within any single substrate, substrate-specific tools remain mature and are not displaced by the scalar.

## 11 Code and Data Availability

This section provides the provenance manifest for every numerical or provenance claim in the paper. Each listed claim traces to a specific code-output artifact at a pinned commit hash in a repository released with this paper. Independent verification is available by cloning the cited repositories at the cited commits, inspecting the cited artifacts, and rerunning cited scripts where the required public datasets and execution environment are available.

### 11.1 Disclosure Note

This paper is authored by an independent researcher without academic institutional affiliation. The empirical work was self-funded and performed across a portfolio of code repositories released with this paper at `github.com/Wise314`. All numerical claims in this paper trace to committed code-output artifacts at pinned commit hashes in those repositories.

### 11.2 External Disclosure Evidence

The chronology of parameter fixation reported in Section 2 is externally anchored by a USPTO provisional patent application filed on 2025-11-20, application number `63/921,348`, with source code from the `system-degradation-framework` repository as the disclosed material. The filing date provides disclosure evidence independent of git history.

## 11.3 Repository URLs

The repositories below provide the paper's parameter-chronology anchor, empirical evidence base, and secondary operational-recurrence artifacts. URLs and role assignments:

- `system-degradation-framework`: `github.com/Wise314/system-degradation-framework` (chronology origin; source material for USPTO provisional 63/921,348).
- `-thermodynamic-stability-prediction`: `github.com/Wise314/-thermodynamic-stability-p` (regime-separation evidence: mechanical, aerospace, electrical, geophysical).
- `quantum-phi-validation`: `github.com/Wise314/quantum-phi-validation` (regime-separation evidence: quantum).
- `phi-controller`: `github.com/Wise314/phi-controller` (regime-separation evidence: computational).
- `llm-phi-stability`: `github.com/Wise314/llm-phi-stability` (substrate-shared structural-role evidence: LLM).
- `phi-bio-stability`: `github.com/Wise314/phi-bio-stability` (scope-boundary evidence: biological cardiac).
- `phi-hybrid-allocation`: `github.com/Wise314/phi-hybrid-allocation` (operational recurrence: routing).
- `phi-controller-quantum`: `github.com/Wise314/phi-controller-quantum` (operational recurrence: control-quantum).
- `phi-objective-controller`: `github.com/Wise314/phi-objective-controller` (operational recurrence: control-general).

## 11.4 Provenance Manifest

Table 6 lists each numerical or provenance claim included in the provenance manifest with its repository, pinned commit hash, canonical output artifact, and manuscript reference. Every row traces to a specific committed file.

| Evidence role / Domain | Repository | Commit | Canonical output artifact | Manuscript ref. |
|---|---|---|---|---|
| Mechanical (bearings) | `-thermodynamic-stability-prediction` | `2e2099c` | `paper_validation/outputs/bearings_components.json` | Sec. 3, Tab. 5 |
| Aerospace (turbofans) | `-thermodynamic-stability-prediction` | `2e2099c` | `paper_validation/outputs/turbofans_components.json` | Sec. 3, Tab. 5 |
| Electrical (grids) | `-thermodynamic-stability-prediction` | `e36dcad` | `paper_validation/outputs/grid_uk_components_log.txt`; `grid_germany_components_log.txt` | Sec. 3, Tab. 5 |
| Geophysical (Donna Lea; foreshock) | `-thermodynamic-stability-prediction` | `7ab6f2a` | `paper_validation/outputs/seismic_*_log.txt` | Sec. 3, Tab. 5 |
| Quantum (qubits, gates) | `quantum-phi-validation` | `21b191e` | `Outputs_MD/MASTER_VALIDATION_OUTPUT.md` | Sec. 3, Tab. 5 |
| Computational (NN kill precision) | `phi-controller` | `77ab987` | `experiments/OUTPUTS.md` | Sec. 3, Tab. 5 |
| Biological cardiac (arrhythmia) | `phi-bio-stability` | `a440b76` | `results/test1_mitdb_cardiac/mitdb_phi_summary.json` | Sec. 3, Sec. 5 |
| Biological cardiac (AFib negative) | `phi-bio-stability` | `5daba22` | `results/test2_afdb_afib/afdb_phi_summary.json` | Sec. 5 |
| LLM (structural-role support) | `llm-phi-stability` | `16fef3d` | `core/phi_llm.py` (lines 45–193) | Sec. 3 |
| Operational recurrence: routing (7 algorithms) | `phi-hybrid-allocation` | `f0d1377` | `Outputs_MD/strict/README.md` | Sec. 6 |
| Operational recurrence: control-quantum | `phi-controller-quantum` | `6676169` | `results/test1_ibm_fez_*.results.json`; `test2_ibm_fez_*.results.json`; `test8_ibm_fez_*.compare.json`; `test9_*.test9_summary.json` | Sec. 6 |
| Operational recurrence: control-general | `phi-objective-controller` | `4a18fc7` | `Neural/results/README.md`; `Neural/Adam_Transfer/results/README.md`; `Quantum/results/README.md`; `Bearings/results/README.md` | Sec. 6 |
| Chronology origin (first fixation) | `system-degradation-framework` | `5ec267f` | `scripts/test_thermodynamic_law.py` | Sec. 2, Tab. 2 |

Table 6: Provenance manifest: each listed numerical or provenance claim in the paper traces to a specific repository, a pinned commit hash, and a canonical output artifact. Repository URLs are listed in the Repository URLs subsection. The USPTO provisional filing on 2025-11-20 (provisional `63/921,348`) provides disclosure evidence independent of git history for the chronology.

## 11.5 Cold Test Artifacts

The preregistered forward test reported in Section 7 is recorded in committed artifacts whose commitment chain is documented by git history: preregistration document `COLD_TEST_PREREGISTRATION.md` at hash `427f2d7` (2026-04-19); cold-test script `cold_test_bearing2_1.py` at hash `2a2bf3f` (2026-04-

20) with source-commit-hash `2e2099c` embedded in the script header; result JSON `cold_test_bearing2_1_result.json` at hash `c521ae6` (2026-04-20); result report `COLD_TEST_RESULT.md` at hash `5da5322` (2026-04-20); amendment `COLD_TEST_AMENDMENT.md` committed 2026-04-22.

### 11.6 Adjacent Prior-Method Identifiers

Source identifiers for the adjacent prior single-substrate methods discussed descriptively in Section 10: US Patent `11,487,030` (seismic-monitoring / earthquake-detection system); US Patent `9,243,985` (metal-fatigue / fracture-fatigue-entropy monitoring system).

### 11.7 Datasets Used

All datasets cited in this paper are publicly available. Public-dataset anchors: the XJTU-SY Bearing Dataset (24); the NASA C-MAPSS Turbofan Engine Degradation Simulation Dataset (3); the MIT-BIH Arrhythmia Database (14); the MIT-BIH Atrial Fibrillation Database (13); USGS strain observations and the USGS earthquake catalog (22); IBM Quantum calibration data across three backends (6); and standard machine-learning benchmarks MNIST (10), CIFAR-10, and CIFAR-100 (9).

### 11.8 Reproducibility

Every numerical claim in this paper traces to a code-output artifact at a pinned commit hash in a repository released with this paper. Reviewers or readers seeking to verify any individual claim may do so by cloning the named repository at the named commit and inspecting the named code-output artifact; where the artifact is script-generated and the dataset is publicly available, the corresponding script can be rerun. The cold test reported in Section 7 is reproducible from the preregistration document and script at the pinned commits; the reported end-of-life value $\Phi = -0.342137$ was obtained from the committed script applied to 491 vibration CSV files from the public XJTU-SY Bearing Dataset under the preregistered protocol.

# Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 63/921,348 (filed November 20, 2025) and additional U.S. provisional patent applications in the portfolio. No license to implement or commercialize the described methods is granted by this publication. 

# Code Availability

The reference implementations, validation test scripts, and audit artifacts are available in the public repositories at `https://github.com/Wise314`. The repositories cited in this paper are listed in Section 11.

# References


[1] S. Aminikhanghahi and D. J. Cook, A survey of methods for time series change point detection, *Knowledge and Information Systems*, vol. 51, no. 2, pp. 339–367, 2017.

[2] V. Chandola, A. Banerjee, and V. Kumar, Anomaly detection: A survey, *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.

[3] A. Saxena, K. Goebel, D. Simon, and N. Eklund, Damage propagation modeling for aircraft engine run-to-failure simulation, in *Proceedings of the International Conference on Prognostics and Health Management*, 2008.

[4] V. Dakos, S. R. Carpenter, W. A. Brock, A. M. Ellison, V. Guttal, A. R. Ives, S. Kefi, V. Livina, D. A. Seekell, E. H. van Nes, and M. Scheffer, Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data, *PLoS ONE*, vol. 7, no. 7, e41010, 2012.

[5] K. He, X. Zhang, S. Ren, and J. Sun, Deep residual learning for image recognition, in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.

[6] IBM Quantum Platform, Calibration data for superconducting quantum processors (accessed across ibm_fez, ibm_torino, and ibm_marrakesh backends), IBM Quantum, 2024–2026.

[7] K. Jamieson and A. Talwalkar, Non-stochastic best arm identification and hyperparameter optimization, in *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pp. 240–248, 2016.

[8] A. K. S. Jardine, D. Lin, and D. Banjevic, A review on machinery diagnostics and prognostics implementing condition-based maintenance, *Mechanical Systems and Signal Processing*, vol. 20, no. 7, pp. 1483–1510, 2006.

[9] A. Krizhevsky, Learning multiple layers of features from tiny images, Technical Report, University of Toronto, 2009.

[10] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, Gradient-based learning applied to document recognition, *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[11] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, Machinery health prognostics: A systematic review from data acquisition to RUL prediction, *Mechanical Systems and Signal Processing*, vol. 104, pp. 799–834, 2018.

[12] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar, Hyperband: A novel bandit-based approach to hyperparameter optimization, *Journal of Machine Learning Research*, vol. 18, pp. 1–52, 2017.

[13] G. B. Moody and R. G. Mark, A new method for detecting atrial fibrillation using R-R intervals, *Computers in Cardiology*, vol. 10, pp. 227–230, 1983. MIT-BIH Atrial Fibrillation Database, PhysioNet.

[14] G. B. Moody and R. G. Mark, The impact of the MIT-BIH Arrhythmia Database, *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001.

[15] P. D. Nation and M. Treinish, Suppressing quantum circuit errors due to system variability, *PRX Quantum*, vol. 4, 010327, 2023.

[16] E. S. Page, Continuous inspection schemes, *Biometrika*, vol. 41, no. 1–2, pp. 100–115, 1954.

[17] G. Pang, C. Shen, L. Cao, and A. van den Hengel, Deep learning for anomaly detection: A review, *ACM Computing Surveys*, vol. 54, no. 2, pp. 1–38, 2021.

[18] L. Prechelt, Early stopping, but when?, in *Neural Networks: Tricks of the Trade*, Springer, pp. 55–69, 1998.

[19] M. Scheffer, J. Bascompte, W. A. Brock, V. Brovkin, S. R. Carpenter, V. Dakos, H. Held, E. H. van Nes, M. Rietkerk, and G. Sugihara, Early-warning signals for critical transitions, *Nature*, vol. 461, pp. 53–59, 2009.

[20] M. Scheffer, S. R. Carpenter, T. M. Lenton, J. Bascompte, W. Brock, V. Dakos, J. van de Koppel, I. A. van de Leemput, S. A. Levin, E. H. van Nes, M. Pascual, and J. Vandermeer, Anticipating critical transitions, *Science*, vol. 338, no. 6105, pp. 344–348, 2012.

[21] C. Truong, L. Oudre, and N. Vayatis, Selective review of offline change point detection methods, *Signal Processing*, vol. 167, 107299, 2020.

[22] U.S. Geological Survey, Earthquake catalog and strainmeter observational data (Donna Lea borehole strainmeter; Tohoku foreshock catalog), USGS Earthquake Hazards Program, accessed 2025–2026.

[23] B. Wang and B. H. Fang, Statistical analysis of the European power grid frequency at high time resolution, *The European Physical Journal Special Topics*, vol. 229, pp. 781–791, 2020. (Data: UK National Grid and German 50Hz grid frequency archives.)

[24] B. Wang, Y. Lei, N. Li, and N. Li, A hybrid prognostics approach for estimating remaining useful life of rolling element bearings, *IEEE Transactions on Reliability*, vol. 69, no. 1, pp. 401–412, 2020. Dataset: XJTU-SY Bearing Dataset, Xi'an Jiaotong University.